

Pricing the Optionality the BuckBasket Sells

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We analyze the Panoptic carry/convexity critique against the BuckBasket and specify the wave-3 markout instrument. We evaluate the basket's own side of the excursion catalogue and record the design decisions reached by working each open question through one historical and one synthetic example.

We map the system as it stands across every design document in the repo, state the theory each decision rests on and the repo text it clarifies or contrasts, give the resulting structure a formal model with falsifiable predictions, and specify the wave-3 program cell by cell. The weak side of the basket's convertibility undertaking is struck as a tranche ladder – successively deeper discounts sized to the reserve remaining – so it has no cliff for an attacker to time. Testability and modelling are the spine: nothing here ships on-chain that has not first won its catalogue cell.

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1 Purpose, goals, and how to read this document

1.1 The question

The Alberta Buck stabilizes BUCK against a defined basket of commodity tokens. Two kinds of flow act on that equilibrium:

- **Organic flow:** BUCK buyers and sellers arriving at varying rates and intensities for their own reasons – savers accumulating and spending, debtors converting mortgages and retiring, depositors entering and leaving the basket, and their waves and runs. Organic behaviour is to be SUPPORTED (deep liquidity at par, cheap execution, no gates) and REWARDED (the basket’s yield against demurrage).
- **Inorganic flow:** agents attempting to disrupt or corner the BUCK/TOKEN markets for profit or malice. Inorganic behaviour is to be DETECTED and HARVESTED: the attacker’s funds transferred to the BuckBasket investors’ ROI.

The basket’s issuance capacity and its reserves from prior profits are to stand ready to amplify that harvest for all position holders – while never holding positions that work AGAINST the restoration of equilibrium (a wall of basket positions that K must push the credit issuers through, or inventory that stalls the recovery it was bought to hasten). Ideally the cost of the basket’s positions is borne by the malicious actor, who must eat through them to recover their funds after disrupting the equilibrium.

1.2 The answer, in one paragraph

The system already holds equilibrium against organic flow and, when a private defender cast exists, against inorganic flow. But the harvest goal is not met, and until 2026-09-01 was unmeasured: at x4 an undefended attacker breaks the peg at zero cost to itself while depositors realize -\$2.0M to -\$2.7M; with the full cast, depositors are spared, the harvest (+\$1.8M realized) goes to private defenders, and the attacker’s -\$10.6M lands on third-party pool LPs and arbs. Harvest accrues to whoever is the COUNTERPARTY at the venue where the attack lands, and the basket is not that counterparty today. The remedy is a ladder target zone: private liquidity harvests day-to-day traffic near par; citizens’ latent credit – unleveraged insured assets behind BuckCredit limits, paying no demurrage while poised – issues into premiums and retires into discounts through an automated facility; and the basket stands behind both as a currency board in bundle units, with an announced UNLIMITED strong-side undertaking (sell BUCK for the bundle at par + eps) and an NAV-bounded weak-side undertaking (buy BUCK for the bundle at par - eps). K sees the SHADOW deviation – bvib plus the inventory the desk and the facility have absorbed or issued – so the slow lever mid-ranges every fast actuator it can see and the wall never blinds the controller. Restoration dominates harvest; the harvest is a small, certain margin on both edges plus a stress fee on duress exits, shared pro rata by a single receipt class; and the real return to investors is the excursion that no longer happens. A bvib-settled perpetual with a funding rate keyed to the shadow deviation charges the DURATION of positions held against the band – the missing long-side toll – and its telemetry sees the accumulation phase. Every element is specified as a catalogue cell with a pass/fail gate before any contract work.

1.3 Reading guide

- Section 2 maps the system as it stands, drawing on every design document, so the rest of the document can refer to mechanisms by name.

- Section 3 states the external critique, where it lands, and the basket-side measurement that reframes the program.
- Section 4 gives the theory: eight results from monetary economics, market microstructure and control, each tied to the repo text it clarifies or contrasts.
- Section 5 records the decisions of record (D1-D7, D9).
- Section 6 draws the resulting structure and its contract surface.
- Section 7 is the model: symbols, laws, bounds, and predictions the catalogue can falsify.
- Section 8 is the wave-3 program, cell by cell, with gates.
- Sections 9-12: risks and open questions, recommendations, glossary, references.

1.4 What this document settles against the other documents

document	what it settles	how this document uses or contrasts it
alberta-buck-ethereum.org	the contracts: BuckCredit, Buck, K, basket, demurrage	the sign convention, the funding factor, the
alberta-buck-ethereum-direct.org	the USD-free embodiment; bvib as process variable	the common-mode axis IS par; why the
alberta-buck-ethereum-basket.org	the heavy anchor; Shannon's demon; arbs work for free	the differential harvest vs the common-m
alberta-buck-rebalance.org / BASKET-REBALANCE.md	the directors: pairs/vrate, leash, persistence, timing	common mode from the same filter bank
BASKET-REDESIGN.md	custody, sell-high redemption, treasury split, regime table	the deflation branch; value conservation;
alberta-buck-operations.org / BuckBasketOps.sol	the desk: quadrants, three bounds, three runaways, the law	D1, D4; the desk is the executor of the u
BuckBasketFence.sol	K-scaled basket issuance; limitation list	D3 splits its concern (persistent stock) fr
alberta-buck-demurrage.org	2%/yr demurrage, Carrying accounts, the Jubilee	D6: poised capital is latent credit, not B
CREDIT-IMPROVEMENT.org	live credit limit, transfer lock, D8 unvetted insurers	latent credit as the wall's ammunition; L
alberta_buck/sim/EQUILIBRIUM.md	the campaign; approaches forward	approach (C) "direct stabilization chann
REBALANCING-EQ.org	the matrix, the catalogue, the roadmap	the instrument; recommendations 2, 4, 5
alberta-buck-ethereum-routing.org	BUCK-unaware arbs route through the basket pools	the benign workforce the treasury seeder
alberta-buck-platform.org	JS port decisions; the sim as specification	only demonstrated agents port; the cata

2 The system as it stands

This section is a map, not a re-derivation; every item names the document that settles it.

2.1 The common-mode loop: par, K, and the funding factor

- **Par is definitional.** In the direct embodiment BUCK measures itself against its basket: $\text{basketValueInBuck (bvib)} = \sum_i \text{basketAmount}_i \times \text{price}_i$ in-BUCK, read as a 600 s TWAP over the basket's own TOKEN/BUCK pools with spot fallback on a cold pool. Setpoint 1.0; error = 1 - bvib (alberta-buck-ethereum-direct.org; master doc, "Direct Variant").
- **Sign convention (decision of record).** bvib > 1: the basket costs more BUCK than it should, BUCK is CHEAP (a DISCOUNT excursion), error < 0, K FALLS, credit contracts, supply shrinks, BUCK recovers. bvib < 1: BUCK is DEAR (a PREMIUM excursion), K RISES, credit expands. Canary tests guard the direction.
- **K is the LTV cap:** $\text{creditLimit} = \text{activatedValue} \times K$. K0 = 0.75, rails (0, 0.95) so a railed K is still solvent; integral-dominant, a sustained 10% deviation spends rail-to-rail authority over $\tau_I = 90$ days with a 2% proportional kick; conditional-integration anti-windup at the rails; bumpless retune and re-prime; dT 1800 s (EQUILIBRIUM.md; BuckKControllerBase.sol). "Deliberately slow: K is a structural lever, not a market maker" (deploy.py).

- **The funding factor** is the on-chain counter-cyclical toll on insured issuance: $ff = \max(0, 1 + 10(b - p)/b)$, 2x at a 10% discount, ZERO once BUCK is 10% dear – issuance is cheapest exactly in a premium. It scales the pre-mint reserve on the insurance principal, not the gross mint; zero-premium credit is exempt. The sim's issue-rate arm ($ff_amp \times rate^{ff_pow}$) has no on-chain counterpart.
- **The theta law** (sim-only, the debtors' decision rule): deploy credit while the all-in execution discount $\leq \theta \times apr$; under an oversupply flood the marginal issuer stops at $\theta \times apr$, so the standing discount is set by the most aggressive unconverted debtor (~8% median, ~19.5% tail) – the flood is untested (REBALANCING-EQ.org recommendation 4).
- **Latent credit** costs nothing until drawn: activation happens only inside `Buck.mint`, premium is charged on present insured value at activation (CREDIT-IMPROVEMENT.org D6), demurrage accrues only on HELD BUCK, and a drawn position "becomes cheaper to close the longer it is carried" through Jubilee relief. D8 – unvetted insurers can write themselves zero-premium credit and mint for free – is the unclosed mainnet blocker.

2.2 The basket: custody, issuance, redemption, treasury

- The BuckBasket registers one TOKEN/BUCK Uniswap V3 pool per constituent and custodies a SINGLE FULL-RANGE Buck-owned LP position in each; it is the sole LP in every basket pool (`deploy.py pool_meta`). Full-range means the CPMM identity $NAV = 2 \times buckReserve$ and a 50/50 value split at any price (BASKET-REDESIGN.md 5.1).
- **DirectMint**: a depositor's TOKEN is paired with basket-minted BUCK at spot – 100% LTV, K-immune (`Buck.mintFromBasket` bypasses credit and funding factor; master doc #10) – and LP'd full-range; the depositor receives an ERC-721 receipt.
- **Redemption is sell-high**: the value claim $\theta \times NAV$ is drawn from the overweight pools by a closed form over the BUCK reserves, valued against a price-scaled fixed-quantity target; value conservation keeps the remaining holders' coverage ratio unchanged. The depositor is paid in TOKEN ONLY; BUCK profit ($Bw - R$) accrues to the treasury; a BUCK shortfall ($Bw < R$, the deflation branch) is covered from the depositor's TOKEN within `maxConversionLossBp` (1%), and $NAV < principal$ reverts (BASKET-REDESIGN.md 5, 6).
- The regime table (BASKET-REDESIGN.md 6) predicts treasury revenue "largest exactly under inflation, when the peg most needs defending" and a costly exit under deflation "so holders keep P outstanding"; the measured version (`plot_basket_split.py`) is that a single agent pushing BUCK 19% dear made every redemption take the deflation branch and cost depositors ~\$1.54M "rather than the agent causing it".
- The directors (`alberta-buck-rebalance.org`): `pairs` (deployed) runs a seven-scale tick-EMA ladder per leg, pairwise quorum 4/7 on the confirmed turn, leash 30/25%, matched-pair trades; `vrate` is the single-window engine; `factor` (trend regime) is Python-only. The COMMON MODE is the mean of the same ladders and was being discarded.
- The ops desk (`alberta-buck-operations.org`; `BuckBasketOps.sol`): the common mode drives four quadrants – Q1 absorb / Q3 supply as self-reversing range orders (repo), Q2 retire / Q4 issue outright under a 30-day-past-the-leash persistence test. Three bounds, each found by a runaway: per-leg 0.4% of the pool's BUCK reserve, position 10% of NAV with inventory-based escalation, cumulative 10% of NAV. The desk is capitalized by `capitalizeMonetary`

("a currency board can only defend with reserves it built up earlier") and may not fund itself from depositor TOKEN. `monetaryDeviationOffset()` exposes what the desk's inventory has taken out of K's error, deliberately not acted on. Default `enabled: false`; it has never run inside a matrix or catalogue arm.

- The fence (`BuckBasketFence.sol`): caps basket issuance at $K \times \text{TOKEN}$ value at TWAP in a $\pm 15\%$ band – "a continuum of resting limit orders" – so that K falling forces withdraw-and-burn and K rising lets it deploy. Its motivating measurement: K at its floor 21% of a 730-day run with the K-immune basket tranche at 47-52% of supply. Limitation 1: the budget is denominated in the unit that moves.

2.3 The excursion program and its instrument

REBALANCING-EQ.org is the design of record. The matrix (ten arms, all PASS at seed 0xA1BC), the raid arms, the portcast cast (settled physics, keyed RNG, no FatCredit; the JS-port cast), and the excursion catalogue: seven injections (none, dump, squeeze, grind-down, grind-up, spike, step) at day 365 of a 2-year window, against defender mixes (usdc = Q1 absorb, buck = Q3 supply, credit = Q4 issue / Q2 retire, basket = two-sided rotation in the bvib-defining pools, crb = the differential rebalancer, all), at defender scales $x1/x2/x4$ and injection budgets $x1/x2/x4$, measured by one response metric (eqmetrics: peak, AUC, recovery, K move and rail days, quadrant flows, P&L). Findings 1-11 stand. The catalogue is the falsifiable instrument this document extends; every cell is a `python -m alberta_buck.sim --experiment catalogue-<arm>.toml --backend pyrevm` run over the real contracts.

3 The critique, where it lands, and the basket's own side of the catalogue

3.1 The critique

Panoptic's "Yin (Carry) & Yang (Convexity) Require Each Other to Survive & Thrive" (X, 2026-08-27; post) is the onchain transplant of the Artemis/Cole framing. In our terms:

A passive AMM LP is a carry trader – a seller of perpetual optionality who collects a fixed fee (the implied volatility) while eating adverse selection (the realized volatility). Onchain, the structural oversupply of LP positions suppresses IV below RV, and crypto is the most momentum-prone, fat-tailed asset class – so the profitable side is BUYING convexity from LPs, not being one. Carry books show negative skew ("sawtooth": steady fees, rare wipeouts); convexity books show positive skew ("stairstep"). A market is complete and stable only when both sides exist and both are PRICED.

3.2 The exposure and the measured evidence

The basket is the sole full-range LP in every constituent pool; its depositor-facing balance sheet IS that LP book; the controller's process variable is read from the same pools; every arb, whale slice and excursion trade in a basket pool has the basket (depositor L + treasury L) as counterparty. Static fee tiers (TOKEN/BUCK 30 bp, BUCK/USDC 5 bp): one implied vol for every regime and counterparty. The depositor is the residual claimant on both commodity moves and par deviation; the treasury holds a one-sided call.

Evidence already in the repo, not previously named: the \$1.54M deflation-branch expropriation (`plot_basket_split.py`); finding 6 (the MA-reversion differential rebalancer loses $\sim 6\%$ /yr trading

macro drift as reversion – $RV > IV$ on the differential axis); finding 11 (foresight turns the same book to +\$4-5M – the informed convexity buyer's profit is real and the quoting side pays it); the ops desk losing on **drift** (2/9 seeds) and the law "a lagged operator is a profitable destabilizer"; the fence test's "genuine impermanent loss that swamps the fee income"; and the absence of any LVR/markout accounting (`snapshot.py` LP APR is gross fee yield).

3.3 The basket's own side of the catalogue (measured 2026-09-01)

No catalogue summary carries a BuckBasket, treasury or depositor column; "attacks punish themselves superlinearly" (finding 8) is a statement about PRIVATE defenders. The frames carry **basketNav**, **treasuryBuck** and **dmProfitUsd**, so the basket side was computed from the raw vectors (units par-USD; end = day 730; deltas only where the mix's pre-history is byte-identical to the undefended cell):

arm	mix	bvib in window	NAV USD end	treasury end	depositor realized end	defender real / marked	whale	K e
dump x1	none	0.99-1.13	47.2M	230k	+0.81M	–	-0.01M	0
dump x1	usdc	0.99-1.09	46.4M (-0.75M)	220k (-9k)	+0.90M (+0.09M)	+0.26 / +0.26	-0.01M	0
dump x1	basket	0.99-1.11	47.6M (+0.43M)	222k (-8k)	+0.97M (+0.16M)	+0.01 / -0.60	-0.01M	0
dump x4	none	0.63-1.12	45.3M	255k	-1.99M	–	0.00M	0.9
dump x4	all	0.95-1.32	50.5M	277k	+0.67M	+1.81 / +3.78	0.00M	0
squeeze x1	none	0.90-1.01	47.8M	223k	+0.94M	–	-0.87M	0
squeeze x1	credit	0.90-1.05	48.0M (+0.16M)	228k (+5k)	+0.89M (-0.05M)	0.00 / +0.63	-0.89M	0
squeeze x4	none	0.69-1.01	47.8M	257k	+0.23M	–	-9.13M	0
squeeze x4	all	0.76-1.04	51.9M	237k	+0.68M	+1.50 / +4.93	-10.59M	0
grind-dn x4	none	0.61-1.02	44.9M	424k	-2.72M	–	0.00M	0.9
grind-dn x4	all	0.97-1.02	49.1M	211k	+0.62M	0.00 / +1.74	0.00M	0

(* railed for the whole tail.)

Four readings, each load-bearing below:

1. **The x4 "dump" and "grind-down" failures are SQUEEZE failures.** The bvib minima (0.63, 0.61) come from the whale's \$30-40M accumulation and re-accumulation legs, which drive BUCK 35-40% dear and rail K at 0.95 – the controller begging for supply the K-scaled issuers cannot deliver. The premium flank is the weak flank, as wave 2 said and as corner theory predicts (4.5).
2. **Undefended malice is free at x4.** The whale ends at \$0.00M in both failed arms while the peg is broken for a year.
3. **Depositors are the residual loser, and the treasury's call pays MORE when the economy is broken** (424k vs ~210k) while depositors realize -\$2.7M.
4. **Defense is a transfer avoided, not a harvest captured.** At x1 the USDC defenders' +\$0.26M coincides with the basket NAV ending -\$0.75M versus the undefended cell: the pools sold the dump cheaply to the defenders. The whale's -\$10.6M at x4 squeeze lands mostly on the BUCK/USDC pool's LPs (SimLP, i.e. third parties) and the arbs, because that is the venue the whale trades in.

The system holds; the basket investor is at best protected and at worst expropriated; the harvest accrues to whoever is the COUNTERPARTY at the venue where the attack lands.

3.4 Where the critique does not land: manufactured reversion

Panoptic's $RV > IV$ inversion is an empirical claim about UNANCHORED, momentum-prone assets. The bvib axis is engineered to mean-revert: K plus the poised cast manufacture the reversion a TradFi option seller merely bets on. Attacker P&L is superlinearly negative on squeezes (-\$0.87M / -\$3.06M / -\$9.13M at x1/x2/x4), defenders are paid countercyclically, and the excursion agents' `min_edge` gate means they refuse to sell convexity below a 1% floor premium. So the two axes split:

axis	vol regime	correct posture
common mode (BUCK vs basket) differential (TOKEN i vs j)	manufactured reversion genuinely trending	SELL convexity – priced, never free; own the edges patience (the director), paid foresight, hedges

This is exactly the split `alberta-buck-ethereum-basket.org` makes between the anchor (the mass that resists being moved) and the harvest (Shannon's demon on reverting constituents): the basket SELLS reversion insurance on the axis it controls and BUYS reversion on the axis the physics controls, and the critique bites only where those two are confused – a per-pool full-range position sells both at once.

3.5 Corrections to v1.0

1. **Reference-price contamination.** Common-mode markout must be measured against PAR (reversion to 1 at horizon D); the truth-implied TOKEN/BUCK reference (via BUCK/USDC) is the pool a BUCK dump moves and serves the differential ledger only.
2. **$\sigma^2/8$ overstates LVR where reversion is manufactured;** realized markout on the common mode, analytic LVR only on the differential.
3. **Symmetric fee surges slow recovery:** only a DIRECTIONAL policy is admissible.
4. **The ops desk was under-credited:** v1.0's options C and F existed as `BuckBasketOps`; the gap is that it never ran in the loop.
5. **The venue was missing:** v1.0 never asked where attacks land.

4 Theory: eight results and the repo text each one clarifies or contrasts

Each result is stated, then tied to the document it explains, sharpens or overturns. These are the load-bearing arguments behind Section 5.

4.1 Carry and convexity: what the basket sells and what it buys

Kyle (1985) and Glosten-Milgrom (1985): a standing quote is a free option to informed traders and a paid service to uninformed ones; the quoter's profit is the uninformed flow's spread minus the informed flow's markout. Milionis, Moallemi, Roughgarden and Zhang (2022) put the AMM case in closed form: loss-versus-rebalancing $\sigma^2/8$ per unit time against a martingale price. Panoptic's contribution is the empirical inversion onchain ($RV > IV$) and the framing that carry and convexity books need each other to be PRICED.

Clarifies `alberta-buck-ethereum-basket.org`: "the world's arbitrageurs, working for free" is true of the BENIGN flow – the uninformed cycles that repin the pools and pay the fee – and false of the informed flow, which the same pools sell the basket's optionality to. The basket doc's yield thesis

(Shannon's demon, +322 to +496 bp/yr measured in `BASKET-REBALANCE.md`) is a DIFFERENTIAL harvest that requires reversion; the COMMON-MODE convexity the same full-range positions sell is a different exposure with a different owner of the reversion (K), and the two must be separated in the position structure. **Contrasts** the routing doc's sim, which measured that BUCK-unaware arbs genuinely route through the basket pools (~1100 routed trades vs ~120 whale snaps) – that flow is the workforce the treasury seeder of D5 exists to lubricate, and it must stay cheap.

4.2 Target zones: a credible band stabilizes inside itself

Krugman (1991): with a credible commitment to intervene at the edges of a band, the exchange rate inside the band is LESS volatile than the fundamentals – the honeymoon S-curve – because speculators front-run the edge. Bertola and Caballero (1992): with IMPERFECT credibility the effect reverses; expected realignment makes the rate more volatile near the edges. The asymmetry is empirical: Hong Kong's strong-side Convertibility Undertaking (unlimited HKD sales at 7.75, 2005) has never been abandoned and never been beaten; the SNB's EUR/CHF floor (2011-2015), mechanically identical, was abandoned because unlimited issuance accumulated a balance sheet approaching GDP in a foreign asset against a FUNDAMENTAL pressure it could not make revert.

Clarifies `BuckBasketFence.sol`: "a continuum of resting limit orders" is a band without announced edges and with a budget "denominated in the unit that moves" (limitation 1); D3 gives the band announced, bundle-denominated edges and keeps the fence for what it is good at – K-scaling the PERSISTENT depositor stock. **Contrasts** the ops doc's reserve doctrine ("any bound it publishes will be probed – the 1992 ERM crisis is the standing lesson"): that is the WEAK side's truth, where the defender spends finite reserves; on the STRONG side the defender issues its own liability, and the correct doctrine is the opposite – announce it unlimited. **Explains** the catalogue: the x4 failures are all on the premium side because the weak side of the current design (private absorbers, the desk's TOKEN book) is reserve- bounded AND the strong side (issuance) is K-scaled and slow.

4.3 The backstop at a penalty rate

Bagehot (1873): in a panic, lend freely, at a penalty rate, on good collateral. The penalty rate is what keeps the backstop from being used in normal times and pays it when it is.

Clarifies the design owner's ruling that "the BuckBasket isn't a market-making liquidity provider, but a stabilizing backstop": the basket's depth sits OUTSIDE the private band, at a small fixed premium eps, and earns that premium for depth-on-demand. **Contrasts** the ops doc's measured trade-off ("stabilization and extraction trade off": 10-day signal peak 1004 bp / desk +\$623k; 80-day 1344 bp / +\$746k; outright inverts past 80 days): the penalty rate is a FIXED small premium, never a stress-scaled extraction, and the recovery is never held hostage (D1).

4.4 Mid-ranging: two actuators, one process variable

Shinskey's valve-position (mid-ranging) control: when a fast small actuator and a slow large one act on the same variable, the fast one controls the process variable and the slow controller does NOT look at the process variable – it looks at the fast actuator's POSITION and drives it back to mid-range. Two controllers on one variable without this separation oscillate or wind up. The Federal Reserve's pre-2008 open-market desk is the monetary instance: the FOMC sets the target; the desk holds the funds rate there with daily temporary repos; ever-larger temporary operations in one direction are the signal for an outright purchase. The FX literature (Jurgensen 1983; Dominguez-Frankel 1993) is the negative result: sterilized intervention – the desk acting while the policy lever is unaware – is weak and transient.

Clarifies `alberta-buck-operations.org` runaway one ("the operator suppresses its own alarm") and `BuckBasketOps.monetaryDeviationOffset` ("exposed rather than acted on, because the right response is a controller-side choice"): the controller-side choice is mid-ranging (D4). **Clarifies** `deploy.py`'s "K is a structural lever, not a market maker; it glides over months while private demand does the fast stabilization" – that is a frequency separation with the coupling left unspecified; D4 specifies it. **Overturms** v1.0's "K-gradient gate" question: the desk and the facility do not gate on dK; their books ARE what K mid-ranges.

4.5 Corners and the ability to deliver

A corner breaks when the cornered asset can be delivered in any size at a known price. The Hunt silver corner (1980) was broken by changing the rules mid-game – effective, and corrosive to trust in the venue. Hong Kong's strong side breaks any HKD corner by standing rule. The SNB shows the one way unlimited delivery fails: when what it accumulates is a liability you cannot make revert.

Clarifies the ops doc's "what only the contract can do – Q4 is not capped by K" and the fence's "the most K-immune half of the float": the K-immune `mintFromBasket` is the system's ONLY anti-corner power, and it must not be blunted by K-scaling; D3 resolves the conflict by duration (persistent stock K-scaled; standing facility K-immune and self-reversing). **Explains** why the basket's unlimited undertaking is SAFER than the SNB's: it sells BUCK dear for TOKEN that is cheap in BUCK, on an axis where par is definitional and K forces the reversion, and Q2 closes the round trip. The only path to the SNB's fate is governance abandoning par.

4.6 The elastic currency and the real-bills doctrine

The Federal Reserve Act (1913) created an elastic currency: money supply expands when banks discount real bills at the window and contracts as the bills self-liquidate. Its historical flaw is procyclicality (credit expands as collateral values rise), which a counter-cyclical lever exists to correct.

Clarifies the design owner's D6 ruling: the poised capital is NOT BUCK but LATENT CREDIT – unleveraged insured assets behind `BuckCredit` limits – issued into premiums and retired into discounts. That is a privately discounted elastic currency with K as the haircut and the funding factor as the discount rate, and the on-chain funding factor already makes issuance cheapest in a premium. **Contrasts** `alberta-buck-demurrage.org` (demurrage punishes idle BUCK) with the demand for standing defense capital: no conflict, because the standing capital is not BUCK. **Sharpens** `EQUILIBRIUM.md` approach forward (C), "a direct TOKEN/BUCK stabilization channel for savers" – the auto-pay-down facility of D6 is that channel, on the credit side.

4.7 Swing pricing: the exiting investor pays for the liquidity they consume

The Reserve Primary Fund broke the buck in September 2008 and those who STAYED bore the loss; the SEC's 2014 and 2023 reforms made liquidity fees on stressed redemptions mandatory for institutional prime funds. Liquity's Stability Pool (opt-in, paid) versus passive LPs in the UST/3CRV pool of May 2022 (conscripted, expropriated) is the same lesson in DeFi.

Clarifies `BASKET-REDESIGN.md`'s regime table – its deflation branch is the pre-2014 money-market fund, and the measured expropriation (\$1.54M; -\$2.7M at x4) is the proof – and the ops doc's "the desk may not fund from depositor TOKEN" (it refuses to conscript explicitly while full-range custody conscripts implicitly). D2's stress fee is the fix the one industry that learned it the hard way adopted.

4.8 Reserve adequacy and the two generations of crisis model

Krugman (1979) / Flood-Garber (1984): a peg defended with finite reserves against a persistent fundamental misalignment fails at a computable date. Obstfeld (1986, 1994): with costly defense, attacks can be self-fulfilling. Both say the same thing about the weak side of a band: its credibility is its reserve relative to the plausible outflow over the horizon on which the slow lever can correct.

Clarifies the ops doc's "reserves are finite and the bound is public" and the catalogue's x4 grind-down with the full cast (parity held, rail margin spent: K 0.936, 17% railed) – that cell is the empirical adequacy frontier. **Sets** the sizing rule of 7.5.

The first-generation model has a constructive corollary: the attack is timed to the DISCONTINUITY – the moment a flat edge backed by a finite reserve stops being honored. Remove the discontinuity and the timing argument has nothing to bite on. A defender whose bid price declines smoothly with its remaining reserve is never "out"; it is only ever "dearer", and the attacker's marginal cost rises with every unit sold into it. This is the shape of every constant-product bid (the catalogue's superlinear squeeze cost is this property of the BUCK/USDC pool), of Curve's stableswap between its flat centre and its convex tails (the amplification parameter A is the dial), and of a central bank that sells reserves by declining-price auction rather than at a fixed rate. **Clarifies** the weak-side design: the design owner's tranche ladder (D3 refinement) is the discrete form, and the flat edge of a classical target zone is its delta $\rightarrow 0$ limit – the one limit with a cliff.

5 Design decisions of record (2026-09-01/03)

Six questions were open after the audit. Each was worked through one historical and one synthetic example with the design owner; the answers are decisions of record.

5.1 D1. Restoration dominates; the backstop earns a fixed premium

Question. When harvest and restoration conflict, which dominates?

Historical. Hong Kong, August 1998: the "double play" (short HKD to force rates up under the currency board; short Hang Seng futures to profit from the crash). The HKMA spent HK\$118bn over ten trading days buying 33 constituents – a bid wall in the venue where the attackers' PROFIT was booked – lifting the index from $\sim 6,700$ to $\sim 7,800$ into the August expiry; the shorts were forced to cover into the wall (harvest at ENTRY). The unwind was restoration-first: held ~ 15 months, then disposed through the Tracker Fund (Nov 1999; HK\$33.3bn first tranche to 180,000 retail investors; $> \text{HK\$}100\text{bn}$ recouped over three years; $\sim \text{HK\$}100\text{bn}$ profit) – from the RECOVERY, into organic demand, never sold back to the attackers at a premium. Then the commitment was hardened (the September 1998 technical measures); no comparable attack since. Sterling 1992 is the contrast: reserves spent against a misaligned fundamental. The difference is whether the defender can MAKE the reversion; par is definitional and K manufactures it.

Synthetic. The ops doc's measurement-scale ladder (above): extraction bought $+\$123\text{k}$ of desk profit for $+340$ bp of peak. A $\sim 20\%$ deviation held a year swung depositors' realized P&L by $\sim \$2.7\text{M}$: the organic cost of prolonging an excursion is of the order of the ENTIRE harvest.

Decision. Restoration dominates. The conflict exists on ONE of four legs:

side	entry leg	exit leg
squeeze (BUCK dear)	Q4 sell issuance at a premium: harvest AND restoration	attacker dumps back $\rightarrow$ Q1/Q2 buy cheap, burn: harvest
dump (BUCK cheap)	Q1 buy the dump at a discount: harvest AND restoration	selling inventory back at a premium: harvest AGAINST r

Harvest is taken on the three coinciding legs. The fourth is unwound TraHK-style – rate-limited into organic flow at par over K’s correction horizon – or burned (Q2) when persistence says the move was real. Extraction is not an objective function.

Refinement. The basket is a stabilizing backstop, not a market maker, and earns a SMALL FIXED premium eps for depth-on-demand (Bagehot). Private LPs inside $\pm x\%$ harvest the traffic; the basket’s deep liquidity sits outside. A recovering dump reaches the basket’s ask side only if it OVERSHOOTS into a premium, so inventory bought on the bid side is sold at a premium only when the market pays it; the remainder is disposed at par. Spot tails (V3 now, V4 directional-fee tails later) are the facility; a perpetual is not (a trader needing depth wants conversion, not a derivative – D5).

5.2 D2. Single receipt class; a stress fee on duress exits

Question. May depositor liquidity be wall ammunition, and who bears the cost among depositors? (Depositors already ARE the wall – involuntarily, unpaid, at every price.)

Historical. Liquity’s Stability Pool (opt-in, paid, held through 2022); the UST/3CRV LPs (conscripted, expropriated); money-market fund reform (the exiting investor pays; a fee is not a gate).

Synthetic. Defended x4 cells show the return to depositors of a paid wall: +\$0.67M instead of -\$1.99M.

Decision. A SINGLE receipt class – every depositor pro rata in band and tails; the harvest (undertaking margins, stress fees, rebalancing premium) shared pro rata. A STRESS FEE on duress exits: a redemption landing in the deflation branch during a measured excursion pays a fee proportional to the deviation, credited to remaining holders. "If you need to exit under duress, you take the hit." Held long enough, K drives bvib back to par and rebalancing garners the reversion premium. Compatible with "always redeemable, no gate".

Caveat of record. "IL does not bite the long-term holder" is true by construction on the common mode and true on the differential mode only to the extent constituents revert – the catalogue’s permanent NRG step leaves a residual. The promise rests on the admission discipline of alberta-buck-ethereum-basket.org ("only civilizational basics"), not on the LP mechanics.

5.3 D3. Two issuance classes; the Convertibility Undertakings

Question. The x4 failures are corners the K-scaled issuers cannot answer; the only actor that can deliver BUCK on demand is the basket via K-immune `mintFromBasket`; the fence wants basket issuance K-scaled.

Historical. Hong Kong’s strong side (unlimited, standing, never beaten); the SNB floor (unlimited, abandoned – the balance sheet it accumulated); the Hunt corner (rules changed mid-game). Standing rules beat emergency ones.

Synthetic. Squeeze x4 undefended: whale -\$9.13M, bvib 0.69, K railed all year. In the isolation runs Q4 fired 2 and 10 times and Q4+Q2 was what the external agent could not do. The fence’s coupling has the SAME sign as the ops quadrants; its concerns – a persistent K-immune STOCK and a budget in the unit that moves – do not apply to self-reversing issuance retired by Q2.

Decision. Two issuance classes, distinguished by DURATION:

1. **Depositor issuance** (persistent stock: DirectMint partner BUCK living as long as the deposit) – K-scaled via the fence once fence-in-the- loop passes its cell, so K governs the stock the economy carries.

2. The Convertibility Undertakings (standing, announced, UNLIMITED in aggregate on the strong side):

- **Strong side.** Anyone may buy BUCK from the basket at $\text{par} + \text{eps}$ paying the basket BUNDLE in `basketAmount` proportions; or at $\text{par} + p \cdot \text{eps}$ paying ANY token with a conversion path known to the basket at deposit time (USDC, BUCK, WBTC, PAXG, ...), the basket spot-converting the incoming asset atomically into its underweight constituents by its rebalancing metrics. p is a convenience factor ($\sim 2 \times$ the route conversion fees; may scale with size for impact). "Unlimited" is the bundle path; the convenience path is bounded by conversion capacity (`maxLegBp` on the route, so the redemption guard never bricks).
- **Weak side.** Anyone may sell BUCK to the basket for the bundle at $\text{par} - \text{eps}$, or for a single token at $\text{par} - p \cdot \text{eps}$, the mix chosen by the basket's under-/over-weight metrics (sell-high) and converted to one TOKEN atomically in the redemption transaction. Reserve-bounded by NAV; the stress fee of D2 composes under duress.
- **The weak side is a LADDER, not an edge** (refinement, 2026-09-02). Because the weak side's reserve is finite and "must outlast K's correction", its bid price scales DOWN with the reserve capacity remaining: a set of positions sized and priced so that liquidity is always available, but successive tranches are available at successively poorer rates. Tranche k is struck at a discount $\text{eps} + (k-1) \cdot \text{delta}$ and sized at a fraction ϕ of the reserve remaining when it is struck, so the reserve depletes geometrically and never reaches zero at a finite price. There is no cliff for an attacker to time (4.8), the attacker's marginal cost rises with cumulative size by construction, and the deviation itself becomes an honest gauge of reserve stress – which is the coupling to K the shadow term wants (7.4). $\text{delta} \rightarrow 0$ is the flat edge; a fixed delta/ϕ ratio is the constant-product bid (the shape the basket's full-range position already has); Curve's stableswap amplification is the interpolation. The same rule applies to the STRONG side's capacity-bounded convenience path (the premium $p \cdot \text{eps}$ rising as the epoch's conversion capacity is consumed); only the unlimited bundle path is flat.
- K-immune by design (fires only at a premium/discount, exactly when the funding factor already says issuance/retirement is cheap); accounted to the monetary book; retired/unwound by Q2/Q1 under the persistence test; bundle-priced so it carries no differential exposure.
- **Announced unlimited.** A published cumulative cap would be probed; what unlimited issuance accumulates here is reversion inventory on a controlled axis, and Q2 guarantees the balance sheet shrinks back.

Consequence. A currency board in bundle units: while both sides are honored, `bvib` is pinned inside $[1/(1+\text{eps}), 1/(1-\text{eps})]$ and Krugman's honeymoon applies inside. The $x4$ squeeze re-run: the pool price cannot exceed $\text{par} + p \cdot \text{eps}$; the whale's \$40M buys BUCK from the basket; the basket books $\sim 2 p \cdot \text{eps}$ over the round trip; the excursion never exceeds the band; K barely moves; depositors never enter the deflation branch. Deterrence replaces harvest – and for basket investors (from $-\$2.7\text{M}$ to $\sim +\$1\text{M}$) deterrence wins.

5.4 D4. K sees the shadow `bvib` (mid-ranging)

Question. The desk absorbs a dump, `bvib` improves, K sees less error and stops tightening, the desk is stranded; under D3 the desk's book is unbounded on the strong side. Sterilized or unsterilized?

Historical. The FX-intervention literature; the Fed desk's temporary- to-outright rule; mid-ranging control (4.4).

Synthetic. Runaway one (49.9% of pool depth, profitable in 7/9 seeds, never escalated); the 90-day run (1.25M BUCK held while K still fell 0.745 -> 0.584 – "not suppressed AT THAT SIZE").

Decision. MID-RANGING. K's process variable becomes the shadow deviation

$$bvib_shadow = bvib + \lambda * (\text{net monetary inventory} / \text{reference depth})$$

with BUCK ABSORBED under the weak side counting positive and BUCK ISSUED under the strong side counting negative – bvib as it would have been without the desk. Equivalently, K is tasked with driving the desk's inventory to zero. Three things follow: the desk's position is never a bet on "K might keep forcing"; Q2 burns need no special treatment (burned inventory leaves the book and the shadow term drops to zero – the temporary/outright and sterilized/unsterilized distinctions are the same distinction); and it IS the achieved-action anti-windup the ops doc asked for. The design owner's statement of intent: "even if we manage to raise bvib back to parity, K must reflect the error we've damped by our action."

5.5 D5. No USD exposure; the treasury seeds BUCK/USDC temporarily; the perp is a funding-rate lever

Question. The attacker's losses land on the BUCK/USDC pool's LPs. Must the basket own that venue, taking BUCK/USD exposure it has no policy control over?

Historical. Counterparty vs venue (COMEX earned fees from the Hunts and harvested nothing; the HKMA stood as counterparty); Hong Kong after 2005 (attacks stopped: deterrence replaces harvest); the cheap band-break bet and the currency board's automatic interest-rate mechanism (overnight HIBOR ~280% in October 1997; the aggregate balance from ~HK\$337bn to ~HK\$45bn in 2022-23 until the shorts gave up), and its 1997 flaw – the rate spike crushed every organic borrower too.

Synthetic. Under D3 the basket owns the EDGES; it need not own the pool. What the undertakings do not do: cap the DURATION of a position against par (inside the band a corner is cheap to hold – a short-BUCK bettor draws credit and pays premium, funding toll and K's contraction; a long-BUCK bettor pays only 2%/yr demurrage; the squeeze flank is weak because holding a squeeze is cheap), or see the ACCUMULATION phase.

Decision.

1. The basket takes NO USD exposure and does not own the BUCK/USDC venue; it becomes a private venue for inside-the-band traffic.
2. **The treasury seeds BUCK/USDC depth TEMPORARILY at launch**, to lubricate the BUCK-unaware arb route that repins the TOKEN/BUCK pools via BUCK/USDC: concentrated positions from the CURRENT price toward the IDEAL (the bundle's USD value at par, read from the truth pools) and not beyond – private LPs supply depth beyond ideal. A single-sided V3 position from current toward ideal is always funded with the asset the CORRECTIVE arb wants to take (BUCK if the price must rise, USDC if it must fall), so it is the counterparty to corrective flow by construction, self-reversing, and withdrawn as private depth arrives – `BuckPoolInvestorAgent`'s placement with a different objective (lubricate, not earn). Orientation is the "silent killer" of the ops doc; verify in sim.
3. **The perp is a bvib FUNDING-RATE LEVER, not a trading venue.** Settled on bvib (par-denominated), never on BUCK/USD; funding set by the system as a function of the shadow deviation; the monetary book as the house, mid-ranged by K. Hong Kong's automatic

interest-rate mechanism TARGETED: it charges positions held against the band and nobody else. Its OI, basis and funding are the accumulation-phase telemetry; its payoff is bounded by the band. Built as a sim agent first, or an installed contract if easily obtained (wave 3f).

5.6 D6. Demurrage stays; poised capital is latent credit; the auto-pay-down facility

Question. Should capital verifiably parked in defense posture be rebated its demurrage?

Historical. Woergl (demurrage's target is the idle hoard); interest on reserves vs negative deposit rates; catastrophe bonds (coupon for committed capital, funded by premiums); the elastic currency (4.6).

Synthetic. Which poised capital EARNED in the catalogue: the USDC absorbers (no demurrage) realized +\$0.26 / +0.28 / +0.39M; every BUCK-holding mix bled; the credit base was scale-stable, +\$0.63M marked, and paid nothing while poised because it holds no BUCK.

Decision (design owner). Demurrage on poised BUCK is not something that should happen – because the poised capital is NOT BUCK. It is LATENT CREDIT: a hopefully revenue-generating insured asset (farms, homes, businesses – the plentiful pent-up un-leveraged asset base) behind a BuckCredit contract with an unused limit, ready for deployment, with no BUCK sitting in an account accruing demurrage. When BUCK is at a premium, holders issue against their limit and buy assets with the temporarily too-valuable BUCK; at a discount they buy BUCK cheap and pay their credit down. Four rules:

1. **No readiness rebate.** Demurrage stays; BUCK parked in the basket earns the single-class harvest (the basket doc's "yield against demurrage" made concrete); weak-side defense capital is TOKEN/USDC/NAV and never paid demurrage; the strong side is issuance, not inventory; the monetary book's own BUCK pays demurrage as an internal transfer to the Jubilee.
2. **eps is LARGER than a backstop alone would choose**, leaving an inner zone where ordinary citizens with available credit "get in on the deflation action" and "buy back their credit limit at a temporary inflation moment". The private latent-credit cast is the middle of the ladder; the basket is the wall behind it.
3. **The auto-pay-down facility.** Automated, user-parameterized machinery attached to a BuckCredit position: issue against the limit when $bvib < 1 - x$ (premium) and place the proceeds (default: a basket deposit under the single receipt class); when $bvib > 1 + y$ (discount), redeem/buy BUCK cheap and repay (retirement happens on the BUY; the burn closes the line). Parameters: x , y , maximum fraction of limit to deploy, target asset. **ExcursionCreditArbAgent's** Q4/Q2 logic as an on-chain facility, pre-priced by the funding factor – an offset mortgage that draws when your currency is dear and repays when it is cheap.
4. **Facility utilization enters K's shadow term (7.3):** deployment in a premium is net issued BUCK, retirement in a discount is net absorbed, exactly like the desk's book. Weak-side extension beyond NAV, if ever wanted, takes the cat-bond form (capped vault, only-toward-par, coupon from realized eps and fee-surge income), never a rebate.

5.7 D7. K mid-ranges the level's POSITIONS; the shadow bvib and the position vector are the two alternatives to test

(Design owner + Claude, 2026-09-03.)

Question. As built, K is one PID on one process variable. Several fast facilities now act on that variable at higher frequency – the desk’s Q1/Q3, the undertakings and their ladder, the facility population, the seeder – and D4 feeds their books back to K through a reconstituted price. Are the facilities themselves P, I or D accumulators of K? Is the whole a higher-level controller, and is it drifting toward an estimator (Kalman) or an optimal controller (LQR / MPC) that should be built explicitly? And is the shadow bvib the right coupling, or should each facility report how over- or under-sold it is, with the metadata a higher controller needs to sum it?

What is built, in control terms. None of the facilities writes K; the agents read buckK only for their own decisions. Each is a BANDED PROPORTIONAL ACTUATOR on bvib – a price band triggers a trade – whose STATE is an inventory $q_i = \text{integral of (absorbed - issued)}$, bounded by a cap. In flow terms that state is an integrator; in price terms, while the facility is unsaturated and the book is linear, q_i / D is the price error the facility removed, so it reads as a proportional reconstruction of the masked error. Saturation is a constraint flag, not a PID quantity. The one coupling to K is the observer’s sum, $bvib + \text{sum_i } \lambda_i q_i / D$, read by the same PID as the price, plus gamma boosting the integral increment when any facility is at a bound.

Expanding that through the PID shows what the shadow term is:

$e_s = e + \text{sum_i } \lambda_i q_i / D$ $K = K_0 + K_p e + K_i \text{Int}(e) + K_d de/dt$ (the price loop, unchanged)

- $\text{sum_i } (\lambda_i / D) [K_p q_i + K_i \text{Int}(q_i) + K_d dq_i/dt]$ (an inventory loop per facility)

So D4 already contains P on position, I on position, and D on position – which is P on FLOW, the disturbance itself measured directly, i.e. feedforward. The terms are right. Three things about the packaging are not:

1. **One scalar λ_i locks the inventory loop’s P:I:D ratio to the price loop’s.** The two loops have different plants. From K to bvib the process is self-regulating (the cast reverts price); from K to inventory it is INTEGRATING – level control – which wants a slow, P-heavy PI (Skogestad’s SIMC rules for an integrating process: $K_c = 1 / (k' \tau_c)$, $\tau_I = 4 \tau_c$). One λ_i cannot tune both. This is the missing lever.
2. **The units drag the reference depth D into the on-chain law.** D is the pools’ BUCK reserve: time-varying, and the very quantity a concentrated position (the seeder’s, a private LP’s) distorts (WAVE3.org decision 8). Written as position over cap, the controller never needs D.
3. **gamma and "pinned vs idle" are symptoms.** With the position loop explicit, a saturated facility’s position pins at its cap and the price error carries the margin automatically; no switching, no schedule. An unreadable NAV is a sensor fault, not saturation (WAVE3.org decision 9, closed 2026-09-03).

Historical. Mid-ranging / valve-position control (Shinskey; Allison and Isaksson 1998; Allison and Ogawa 2003; Haugwitz, Karlsson, Velut and Hagander 2005 on anti-windup in mid-ranging), already in 4.4. HABITUATING CONTROL (Henson, Ogunnaike and Schwaber 1995): two inputs, one output; the fast expensive input takes the transients, the slow cheap one takes the steady-state load, and the cost asymmetry is explicit in the objective – that asymmetry IS the carry critique (Friedman 1951’s carry-cost objection to the commodity-reserve currency; the habituating answer is to keep the stock near zero by letting the cheap lever do the persistent work). AVERAGING LEVEL CONTROL (Cheung and Luyben 1979; McDonald, McAvoy and Tits 1986): inventory is a surge tank, K the outflow valve, net demand the inflow; the objective is the smoothest outflow

that never overflows – "K glides" plus the capacity constraint – and its optimal policy is a small MPC. The INTERNAL MODEL PRINCIPLE (Francis and Wonham 1976): a persistent demand shift is a step into the inventory integrator, so zero steady-state inventory REQUIRES an integrator on inventory; the I term on position is not optional. Skogestad (2000, 2004) on plantwide control-structure design: the levels table of 6.4 is the standard regulatory / supervisory / optimization layering, and "aggregate position = 0" is a self-optimizing controlled variable (hold it, and carry is minimized without knowing the disturbance). In market microstructure the shadow bvib formula is the Avellaneda and Stoikov (2008) reservation price, mid + inventory x risk aversion – a market maker's OWN quoting variable, a level-1 object. The monetary instances: HKMA's Convertibility Undertakings are level 1 with no level 3, and the aggregate balance swings accordingly; the Fed's temporary-to-outright rule is level 2; sterilized intervention (Jurgensen 1983) is the negative result.

Optimal control and estimation. The reduced plant (7.2) is linear with a handful of states: price error, aggregate position, their integrals, K. With costs on price error squared, position squared weighted by carry, and the change in K squared, LQR returns a linear feedback on exactly the vector [e, Int(e), s, ds/dt] – "each facility's signal, with metadata, summed by a higher controller" – with the weights DERIVED from stated costs instead of swept, and it derives the lambda knee D4 asserts. LQR ignores constraints; MPC handles them, but a QP per `compute()` is not an on-chain law. Explicit MPC for a box-constrained system this small is piecewise-affine (Bemporad et al. 2002): a linear law inside the operating region plus rail logic – which is what a mid-ranging cascade with anti-windup already is. So: optimal control EXPLICITLY IN THE DESIGN, offline in the sim (WP-16), and a fixed linear state feedback with rails on-chain. Take gain RATIOS from LQR and absolute gains from stability margins, because the elasticity from K to flow is the least-known parameter. A Kalman filter is not the tool: inventory is measured exactly and the observer is algebraic. The one hidden variable is the PERSISTENCE of the demand flow, which decides temporary versus outright (level 2); that is a disturbance observer (Chen et al. 2016) – a low-pass on the position rate with a threshold and a dwell – and it is the ops doc's persistence test made explicit. The price gauge is the one noisy sensor, and its fix is decision 8, deterministic. The K loop's bandwidth is bounded by its dead time through credit issuance; the facilities exist to cover that window, and their capacity must cover dead time x worst-case flow (7.4's reserve-adequacy sum).

Decision (design owner). BOTH designs are kept as alternatives, to be implemented and tested on the same grid (WAVE3.org, "The controller-alternatives test plan"); the missing levers are corrected in both.

The corrected controller (one implementation, two aggregation modes). K runs TWO loops on one lever, one integrator each:

$K = K_0 + [K_p e + K_i \text{Int}(e) + K_d de/dt]$ the price loop (Direct)

- $[K_q s + K_{qi} \text{Int}(s) + K_{qd} ds/dt]$ the position loop

with rails and per-integrator anti-windup, s the observer's aggregate position, and (Kq, Kqi, Kqd) INDEPENDENT of (Kp, Ki, Kd). The observer offers two aggregations of the same **IStabilizer** books:

- **S (shadow, D4 in its own units):** $s_S = \sum_i \lambda_i q_i / D$, dimensionless, in price units. With (Kq, Kqi, Kqd) = (Kp, Ki, Kd) the corrected controller reproduces D4 as built, term for term in the interior (at a rail the anti-windup is per integrator, so the identity is interior-exact and rail-approximate); with independent gains it is D4 with the missing lever.
- **V (position vector):** $s_V = \sum_i w_i q_i / \text{cap}_i$, with cap_i the facility's own inventory

bound in BUCK and w_i a governance COST WEIGHT (sum $w_i = 1$), so s_V is the level's cost-weighted fill in $[-1, 1]$, the same units as saturation, and D never enters the law.

Facilities do NOT declare a P , D or I type: every facility reports the same physical state, and the controller derives position, its integral and its rate, as it does for price. The metadata a facility owns is its cap, its saturation, and its cost weight. `IStabilizer` gains `position()` (q_i / cap_i , signed 1e18, absorbed positive), `positionCap()` (cap_i in BUCK) and `stale()`; `netInventory`, `capacity` and `saturation` stay. γ is retained as an optional schedule with default 0: with the position loop explicit its job is done by structure, and it is measured, not assumed. With $K_q = K_{qi} = K_{qd} = 0$ (and $\gamma = 0$) the controller IS `BuckKControllerDirect`; that identity replaces the $\lambda = \gamma = 0$ identity as the regression test. The price loop and `fundingFactor()` read the RAW bvib (WAVE3.org decision 10, 2026-09-04): the position enters K only through the position loop, and participants learn of the interventions through K 's motion, not through the mint gate.

The lever count. Integral action can zero at most as many controlled variables as there are independent manipulated variables. K is one lever; it can zero ONE position – the aggregate s – so the controller carries one position integrator, never one per facility (two integrators on one lever with different residuals wind against each other). The DISTRIBUTION of inventory among the facilities (the differential modes) therefore needs its own levers, and today it has none: no facility skews its own band by its own inventory. Each level-1 stabilizer gets a LOCAL SKEW,

$$\text{band}_i(t) = \text{band}_{i0} + \kappa_i q_i / \text{cap}_i$$

the Avellaneda-Stoikov reservation-price adjustment applied to the facility's own edge (the desk's Q1/Q3 thresholds, the undertakings' eps on each side, the facility's x / y , the seeder's range centre): a facility long inventory bids lower and offers sooner, so its own flow reverts its own book. This is where the shadow-price idea belongs – one level down. κ_i is a per-facility P (or slow PI) on its own position; K 's aggregate integrator is the common mode. Common mode to K , differential modes to the local skews: n positions, n levers (WP-14).

Sensor faults. A facility whose cap cannot be read reports its LAST GOOD cap with a staleness flag; position is always readable (`netInventory` depends on nothing external); saturation follows from the held cap; a disabled stabilizer is excluded from the aggregate. Never "pinned", never "idle" (WAVE3.org decision 9, where the mechanics of an unreadable NAV are recorded).

What is NOT done. No on-chain MPC or QP; no Kalman filter on inventory; no per-facility integrators in K ; no reconstituted price as K 's input under V (S keeps it, as the alternative under test); no fast-controller filtered signals into K (6.4 stands).

Optimal control, as derived (WP-16, 2026-09-12). The LQR of record is LOOP-AT-A-TIME. With one lever and both integrals costed, the reduced plant has a conserved direction – k_s (a $\text{Int}(e) + e$) + $k_e s$ is unchanged by any input, so the price integral and the position carry the same information (the impact-scaled cumulative net flow) and cannot both be zeroed by K – and the joint Riccati iteration diverges. Each loop's LQR on its own plant reproduces the sim's price gains exactly (the anchor) and gives the position loop K_q 2.42 per unit s_S , K_{qi} 8.1e-8/s, τ_s 127 d at the carry weight of record; the joint closed loop then carries a structural eigenvalue at 1 (the two integrators' contents can shift between them without moving K), bounded in the sim only by the band and the rails. A joint law would need the band – a hybrid, which the cascade with anti-windup already is. Two facts of the sim shape what those gains can do: K is a weak lever ($\sim 50k$ BUCK/day of flow per unit), so the derived position loop spends its whole rail at a 21% fill of the undertakings' reserve and the $S0$ -equivalent λ is ~ 24 against the 0.5-1 the D4 cells ran; and the undertakings' book on a calm arm is transient, unwound at par within days. WAVE3.org decisions 27-31 carry the choices this leaves: the persistence horizon, the position loop's authority,

the depth axis, P9's window, the two-integrator drift. What follows from these findings for the structure itself is D9 (2026-09-13).

What decides between S and V. The grid: the same disruptions, the same facility mixes and capacities, both aggregations, gains derived the same way (WP-16), judged by G1-G8 plus the habituation and book-loading gates (predictions 9 and 10). V is expected to win on units (no D), on the guard-trip days, and on tuning transparency; S is the incumbent and may win on simplicity if the position loop's gains turn out to want the price loop's ratios anyway. R5 is amended: K's input is the observer's aggregate position in EITHER units, with its own gains.

5.8 D9. K takes ONE integrated signal; the flow's persistence is the hidden state; the sequence from here

(Design owner + Claude, 2026-09-13. D8 is CREDIT-IMPROVEMENT.org's unvetted-insurer decision, referenced throughout this document; the number is skipped here so that reference stays unambiguous.)

Question. WP-16 (reports/wp-16.org) derived D7's position-loop gains and, on the way, showed that the joint plant is not stabilizable: one lever, two costed integrals, a conserved direction. The design owner's reading: the theory was that K could be moved slowly on persistent price error (the I term on bvib) and more quickly on the scale of the monetary operations the fast events required (the position loop), but both inputs measure the same thing, so they fight – one gaining while the other damps, by the timing of their measurements and responses. Two ways forward were put: (a) back to a single PID, with the short-horizon events synthesized into its P and D inputs and larger K_p / K_d so that K answers faster; (b) deeper measurement of the economy – the scale and dynamics of what moves bvib, not bvib alone – feeding an optimal-control law that computes the K which would soft-land bvib at par if present trends continued, re-planned as they do not, the way a landing rocket estimates the wind from its own divergence and feeds that estimate forward. This section records what WP-16 measured, what it means in control terms, which of the two ways is right and why, and the sequence of experiments that decides the design from here – for each stage, what must be discovered, why it matters, and the design each outcome selects.

What WP-16 measured. Findings 1-9 of the report; cells build/sim/kdesign/d10-{ctrl,m20,m10,m05,p05,p10,d40-{ctrl,m20,p20} and the grind set (1-year cells, K held 0.75 from day 120, the K0 step at day 150). Every number is from the thin D2-world cast and provisional under ORGANIC-SCALE.org decision 6.

quantity	measured	finding
supply flow per unit K	~50k BUCK/day, as a RAMP; 56k on a loaded book	1, grind
at full rail authority (dk_rail 0.5)	~25k BUCK/day	1, 5
a 60-day grind of \$8-12M	130-200k BUCK/day	2, grind
draining a full \$15M book at the rail	~600 days	5
price shift from full authority	0.2-0.4% of bvib	3
daily price error, loop open	std 0.70%, lag-1 autocorrelation -0.53	3
a cut to K 0.55	-6.3M BUCK at once (the debtors' limits bind)	1
the undertakings' book on the calm arm	transient: unwound at par within days	2
the derived position loop's rail	reached at a 21% fill of the real reserve	5
the two integrators' joint eigenvalue	exactly 1, eigenvector (0, Kqi, 0, -Ki)	4
the LQR's flow-state gain (the feedforward)	$L_d = 210 K$ per (s / day); unrealisable as a D term	9

The finding in control terms. The design owner's reading is right, and the conserved direction states it exactly. The price integral and the facilities' inventory are two GAUGES on one quantity: the cumulative net demand for BUCK at the peg. K moves supply, and the band routes that supply into one gauge or the other. While a facility holds price the flow lands in inventory and the price gauge reads nothing; when the facility is drained or absent the same flow lands in price and the

inventory gauge is pinned at its cap. K never has two independent errors to serve. It has one error read by two instruments, and which instrument is live depends on the regime. 4.4 already said so: "the slow controller does NOT look at the process variable – it looks at the fast actuator's POSITION ... Two controllers on one variable without this separation oscillate or wind up." D7's corrected controller put BOTH loops on K, and WP-16 measured the consequence.

The fight has a specific shape. A facility absorbs a lump and unwinds it at par within days (finding 2). The position integrator keeps the residue of that excursion; the residue holds K slightly low; the standing price error that leaves is what the price integrator then winds against, on its 90-day constant. The two integrators end up holding offsetting contents whose sum is the K the plant wants, and the split between them drifts freely – the eigenvalue at exactly 1 (finding 4). In the contract only the rails bound it, and the worst case is both integrators pinned at opposite rails with K interior, the per-integrator anti-windup fighting a saturation that is not real.

Why faster K gains are not the answer (way (a)). Three measured facts make K a slow lever by physics, not by tuning:

1. The daily price error is noise on the day scale (std 0.70%, lag-1 -0.53): the cast restores the price inside the day and overshoots. A larger K_p moves K by that noise; a K_d amplifies it. Finding 9 is the derivative form: the implied flow gain would jerk K by tenths from one lumpy fill.
2. K acts through the borrowers' arrival hazard and their K-scaled limits (`equilibrium_agents.py`, the k / k_ref hazard). A step in K yields a supply RAMP that compounds late (+8k BUCK at day 240, +2.15M at day 364 for +0.20). A K move lasting a day moves supply by a few thousand BUCK. No gain makes a one-day K move matter.
3. The lever is asymmetric. A cut to 0.55 delevered 6.3M BUCK at once – a margin call; a rise is an offer nobody has to take (cuts 55k vs rises 44k BUCK per K-day at depth 10; 44k vs 75k at depth 40). A linear controller of any speed does not know this.

Short-horizon events belong to the level-1 facilities by physics: they hold inventory, K does not. What CAN be faster is not K's response but the DECISION about which part of the flow is K's business – the persistent part. Way (b) delivers exactly that, and it is a different mechanism from higher gains. Way (a)'s single-integrator half is right and is kept (stage 1); its higher-gain half is not. This also settles WAVE3.org decision 7 in the direction WP-16's anchor found: the 2% P term is what a quadratic cost on the instantaneous error wants when the cast itself reverts within the day, and no larger P is justified by any measurement in hand.

What way (b) is, in this system's terms. D7's paragraph "Optimal control and estimation" named the one hidden variable, the PERSISTENCE of the demand flow, and its tool, a disturbance observer. WP-16 then derived the law that wants it: the LQR's flow-state gain $L_d = 210$ says the optimal K is MOSTLY FEEDFORWARD from the disturbance flow, and finding 9 says it was unrealisable only because the flow was taken by differencing one lumpy gauge. The missing component is an ESTIMATOR that fuses every channel through which excess demand is met – the facilities' fills, the desk's net inventory, direct-mint net flow, debtor draws and repayments, the pools' reserves – all on chain already, so the sensor is free. With the disturbance modelled as a persistent component (decay T_p) plus a transient one (decay τ_t , days), the steady-state Kalman filter is an alpha-beta tracker on the fused flow: two exponential averages in the contract (Kalman 1960; Kalata 1984). The persistent estimate lags by $\sim \tau_t$, not by the 90-day price integral. The law:

$e_s = e + \lambda s$ the shadow error: ONE signal (S units; in V units s_V / λ_V)
 F_hat = the fused net flow (BUCK/day, sold positive); F_p its persistent part (a slow average);
 $F_t = F_hat - F_p$ $K = K_0 + \text{clip}(-F_p / g, \text{rails})$ feedforward: the K whose supply ramp cancels the persistent flow

- $K_p e_s + K_i \text{Int}(e_s)$ ONE PI – the price loop’s – on the shadow error (S0k as built)

with g the supply ramp per unit K (WP-16’s F_K , identified, not assumed) and the one integrator covering only the part of g we got wrong – the internal model principle (4.6) doing the one job it is needed for. The landing: K leaves the rail when the supply still in the pipe – g times the uptake lag τ_K times the K excursion – equals the book remaining; that is the powered-descent timing of the design owner’s analogy (Acikmese and Ploen 2007), and P9’s one-sign-change test measures it. The price loop on K is not a second loop: the price term of the shadow error covers the regime in which no facility is engaged, the position term the regime in which one is. The second lever is the FACILITY (its inventory and its local skew, D7); the third, where it exists, the desk’s outright operations.

Two limits on the analogy, of record. The rocket knows its physics to a percent and the wind is a perturbation; our g differs 2x between cuts and rises and depends on the cast, and a dump is 5-8x K ’s authority. So during a disruption the feedforward sits at the rail, the facilities absorb, and the estimator earns its keep on the RELEASE. And every number above is from the thin cast: the ratio of K ’s authority to the disturbance scale is an assumption about how much of the money supply is credit-created, and belongs on the organic ladder (ORGANIC-SCALE.org) as a headline number per rung.

The structural conclusion. K takes ONE integrated signal. The second lever is the facility. The sensor is an estimator of the flow and its persistence – level 2 of 6.4 made explicit: the ops doc’s persistence test as a filter instead of a dwell. D7’s S and V stay under test as the two-integrator alternative, and cycle 1 (in flight) is read as the single-versus-double integrator test it already contains. Proposed for the design owner (WAVE3.org decision 38): the two-loop form of D7 becomes the alternative and one PI on the shadow error the incumbent once stage 1 reads that way; the estimator follows stages 2-3.

Recommended sequence (Claude, 2026-09-13; for the design owner). Four stages, each cheaper than the next and each bounding the value of the next. For each: what must be discovered, why it matters, and the design each outcome selects.

Stage 1 – cycle 1 as the single-versus-double integrator test. Already running (WAVE3.org, d7-c1): designs A (price only), S0 and S0k (ONE integrator on the shadow error, λ 1 and the derived 24), S and V (two integrators), V_k (V plus the local skews), on none / dump x1 / dump x4 / cohort-step / guardtrip / bookload.

- To discover: whether the second integrator hurts. Measurements: the gates G1-G8; P9’s sign changes and habituation time; the rail days; and the DRIFT, the total variation of $\text{pid}_i - \text{pid}_{qi}$ (WAVE3.org decision 31; both terms are in every frame, so it is computed from the vectors after the fact in *eqmetrics*).
- Why it matters: it decides whether D7’s two-loop form survives at all, before any estimator is built and at no extra compute.

stage-1 outcome	design selected
S0k matches or beats S and V on the gates, with less drift and fewer sign changes	retire the second integrator: the controller of record is one
S or V beat S0k	the derived λ mapping is wrong, or the position P te
neither S0k nor S / V beats A on the gates	K’s tuning is second order on this cast; go to stage 2 to bo
drift large (an integrator at a rail with K interior on any arm)	the anti-windup review of decision 31 is mandatory whiche

Stage 2 – the perfect-measurement benchmark (the ceiling). Way (b) with the sim’s TRUE flows fed to the controller and g known: a Python controller in the harness (the shadow controller’s PI plus the feedforward with the release rule, its inputs read from the deployment’s own books and the whale’s schedule rather than estimated), one 2-year cell per cycle-1 arm.

- To discover: the value of K on this cast, as a ceiling: the gap between the perfect-measurement K and A , and between it and $S0k$, on the dump, grind and cohort-step arms (peak, AUC, the book’s BUCK-days, rail days, K ’s total variation, habituation time).
- Why it matters: it bounds every later stage. No estimator beats a controller that knows the flow; if that controller barely beats A , the position loop’s authority (decision 28) answers itself and the program’s spend moves to the facilities and the organic economy.

stage-2 outcome	design selected
ceiling within the gates’ noise of A	K is not the lever for these disturbances: decision 28 (b);
ceiling beats $S0k$ by more than the noise on peak, AUC or BUCK-days	the feedforward is worth building: stage 3
ceiling within the noise of $S0k$, both well above A	the single integrator already captures K ’s value; stop at st
ceiling beats A only on the release (fewer sign changes, shorter dwell), not on peak	build the estimator for the release rule only; the feedforwa

Stage 3 – the estimator and the feedforward, in the sim. The alpha-beta estimator on the fused flow with the persistent / transient split, the law above, g and τ_K from identification. Contract form last, and only if the sim form wins its cells.

- To discover: (i) g and τ_K separately for cuts and rises, from 2-year step tests (the 1-year cells showed the ramp still compounding at day 364; whether it saturates or grows decides whether K is a RATE lever or a GROWTH lever); (ii) the estimator’s lag and its false-persistence rate on the control arm (how often organic noise is called persistent); (iii) the release timing against the ceiling ($P9$ ’s one sign change); (iv) the on-chain cost (two EWMA’s and a division against the tick budget).
- Why it matters: it is the design the design owner asked for, and its gains are derived from identified constants, not swept, so its transfer across the ladder’s rungs is a matter of re-identification.

stage-3 outcome	design selected
estimator within the gates’ noise of the ceiling on every arm	the contract form is two EWMA’s and the law above; a WP replaces the pos
the persistence call lags beyond τ_t , or fires on organic noise	the disturbance model is wrong: label the flow by source (debtor / direct-m
g asymmetric beyond $2x$	a piecewise law: cuts costed as impulses (the margin call), rises as a rate; ne
the ramp compounds (no saturation inside 2 years)	K is a GROWTH lever on the debtor population: the plant is population dy

Stage 4 – multi-lever optimal control. K , the desk and the facilities as one allocation problem with the same estimator as sensor: the transient component sized to the desk’s operations, the persistent to K , the differential modes to the local skews. That plant has n levers and IS stabilizable; it is habituating control (Henson et al. 1995) done properly.

- To discover: the authority ratio per rung (K ’s flow per unit against the disturbance scale) and the desk’s transient share.
- Why it matters: by the design owner’s thesis and ORGANIC-SCALE.org decision 6 it waits for the organic economy at rung 2; built on the thin cast it would be judged on numbers the cast does not have.
- Outcome: the allocation rule of the monetary-operations program; not designed here.

What this leaves unchanged. The undertakings, the ladder, the stress fee, the local skews, the **IStabilizer** seam (position, cap, staleness; the estimator reads the same books plus the flows the contracts already emit), decisions 9-11 and the organic ladder. What it changes: the two-loop form of D7 is now the alternative under test rather than the incumbent, pending stage 1; the estimator is the design of the next two stages; and "faster K" is closed as a direction. *Ruling (design owner, 2026-09-13).* The analysis and the sequence accepted, with THE STAND FIRST: the two numbers WP-16 rested on (g as the **endog** arrival hazard; the within-day reversion as the arbs' parameters) are population assumptions, so stage 1 runs now, stage 2 rides the organic ladder as a per-rung instrument, stage 3 waits for rung 2 to pass its gates, and stage 4 follows. The record of the stand – the decision kernel, the roster order, the gates, the progression cell – is **ORGANIC-SCALE.org** "The stand before the controller"; the phase rule is in **WAVE3.org** STATUS.

6 The resulting structure

6.1 The ladder target zone

bvib (basket in BUCK) who acts, and how

$> 1/(1-\text{eps})$ sloped, no cliff **WEAK-SIDE LADDER**: the basket buys BUCK for the bundle at par - eps, then at successively deeper discounts as its reserve depletes (tranche phi of the remaining reserve per delta step; per-leg maxLegBp) $1+y \dots 1/(1-\text{eps})$ discount zone latent credit **RETIRE**S (the facility); private Q1 absorbers; private LPs' tails; the desk's Q1 range orders $1-x \dots 1+y$ the band private LPs harvest traffic; the treasury seeder (BUCK/USDC, launch only); the honeymoon effect $1/(1+\text{eps}) \dots 1-x$ premium zone latent credit **ISSUES** (the facility); private Q3 suppliers; the desk's $Q3 < 1/(1+\text{eps})$ cannot persist **STRONG-SIDE UNDERTAKING**: the basket sells BUCK for the bundle at par + eps (**UNLIMITED**; convenience path par + $p*\text{eps}$, bounded by conversion capacity)

across all zones: K mid-ranges the desk + facility books (bvib_shadow); the bvib perp charges the **DURATION** of positions held against the band; the directional dynamic fee prices raid-speed displacement at tick speed; the stress fee prices duress exits.

Deterrence is the product; harvest is the by-product; both accrue to the single receipt class pro rata.

6.2 Where each flow lands, and who is the counterparty

flow lands in counterparty harvest

organic small (savers, the band: private LPs, private LPs fees (theirs) debtors, DM churn) BUCK/USDC, **TOKEN/BUCK** organic large (a refinance, the band then the zone private + facility par+/-x, cheap a depositor wave) then, if needed, the edge then the basket eps (basket) inorganic squeeze strong-side undertaking **THE BASKET** $p*\text{eps}$ entry + (once past par + $p*\text{eps}$) $p*\text{eps}$ exit inorganic dump weak-side ladder **THE BASKET** (NAV) $p*\text{eps}+$ entry, (once past par - $p*\text{eps}$) + private Q1 organic unwind inorganic accumulation the perp (OI/funding) and the house (monetary funding (quiet phase) the band book) x duration raid-speed block the directional fee at the LP position surge fee tick speed (single class) duress exit the deflation branch remaining holders stress fee

6.3 The contract and agent surface (new and changed)

element	where	change
strong-side undertaking	BuckBasketOps (or a sibling shell)	buyBuck(bundle) at par+eps; buyBuck(token)
weak-side ladder	same	sellBuck(bundle) at the current tranche's di
stabilizer seam + observer	IStabilizer (new); BuckBasketOps (the observer, a facet later)	IStabilizer {netInventory, capacity, saturati
controller input	BuckKControllerShadow	_readReferences reads the observer; gain sch
persistence-driven retirement	PairsRebalanceDirector / BuckBasketOps	existing Q2/Q1 escalation applied to the und
stress fee	BuckBasketProRata._redeem	deflation-branch fee ~ deviation, credited to m
the facility	new contract, user-parameterized	draws against a BuckCredit line into a basket
treasury seeder	sim agent, then treasury program	BuckPoolInvestorAgent variant: current -> i
the perp / house	sim agent + Python venue ledger, then contract	funding = g(bvib_shadow); OI cap; house =
directional dynamic fee	sim pool adapter, then a v4 hook behind the venue seam	base fee on the corrective side; RV- and size-s

6.4 The hierarchical structure: the stabilizer seam and the monetary Diamond

(Design owner + Claude, 2026-09-02.) Once several fast actuators act on bvib – the pools, the desk, the undertakings and their ladder, the facility population, the house – "which contract holds K" stops being a question about addresses and becomes a question about signals. The rule from mid-ranging control (4.4) settles it:

K consumes the fast actuators' POSITIONS and their SATURATION, never their filtered signals.

6.4.1 The two seams, and which one is artificial

Buck holds two authorities: **buckK** (an **immutable**; the **IBuckK** surface **currentBuckK** / **compute** / **fundingFactor**) and **basket** (one-shot). The **IBuckK** seam is NOT artificial: it is what lets the controller be swapped and forge-tested as the mathematical specification without touching **Buck**. What has become artificial is the ADDRESS behind it. **BuckKControllerDirect._readReferences** reads one number from one contract; under this design the process variable is assembled from several books that all live on the basket side. The observer that assembles them belongs where the state lives – the monetary Diamond – and **Buck**'s two authorities should both point at it. Because **buckK** is immutable, that address must be the Diamond (or an upgradeable controller proxy) from the day **Buck** is deployed; the decision is recorded in **WAVE3.org**.

6.4.2 The levels

level	timescale	who	controls
0	tick	the pools; the directional fee hook	the price
1	hours-days	the undertakings + ladder; the desk's Q1/Q3; the facility population; the house	bvib, by t
2	weeks	the persistence test -> Q2 burn / Q4 acknowledge	converts s
3	months	K	the stock
4	governance	eps, p, delta, phi, lambda_i, gamma, gains, rails	time-locke

(D9, 2026-09-13.) Level 2 is where D9's flow estimator lives: the persistence test as a FILTER on the fused flow through every level-1 book, feeding K's feedforward, instead of a dwell on a standing book. Level 3's K then integrates ONE signal, the shadow error.

6.4.3 The stabilizer seam

Every level-1 actuator implements one interface:

interface IStabilizer { function netInventory() external view returns (int256); / *absorbed - issued, BUCK (6-dec) function capacity() external view returns (uint256); / remaining, 1e18 = all: rho (ladder), / C_leg room (strong-side convenience / path), headroom (facility), OI room (house) function saturation() external view returns (uint256); // 0..1e18, degree at a bound }*

The OBSERVER registers stabilizers with governance weights λ_i and computes

$\text{shadowValueInBuck}() = \text{bvib} + \sum_i \lambda_i \text{netInventory}_i / D$ $\text{saturation}() = \text{weighted max}_i \text{saturation}_i$

The CONTROLLER's `_readReferences` reads the observer instead of the basket, integrates the shadow error exactly as today (rails, conditional integration, priming, dS , dT), and uses `saturation()` for bounded gain scheduling – $K_{i_eff} = K_i (1 + \gamma \text{saturation})$, γ governance-set and swept – so that when the fast actuators are pinned at their bounds the structural lever moves faster. That is the ops doc's "escalate on inventory, because inventory is the one signal the operator's own action cannot suppress", generalized to the whole level. With every $\lambda_i = 0$ and $\gamma = 0$ the controller IS `BuckKControllerDirect`; that identity is the regression test.

Three things the structure deliberately does NOT do, each for a reason already measured:

- **It does not feed the fast controllers' filtered deviations into K.** The desk's 20-day common mode is a filtered bvib, which K already integrates; feeding both double-counts, and the ops doc's ladder scan showed a lagged reading INVERTS the outright quadrants past 80 days. Positions carry the information without the lag.
- **It does not let the fast controllers set K, and K does not set their setpoints.** Cascade control is the wrong topology: the fast controllers' setpoint is par, which is definitional; K mid-ranges THEM.
- **It needs no priority logic among the fast controllers.** The ladder target zone (6.1) is already a split-range structure: each actuator's active zone is a band of bvib. The one real coupling is shared pool depth (the ops doc's "the desk and the rebalancer compete for the same actuator"), which the observer resolves with a per-epoch capacity budget (C_{leg} accounting), not with priorities. The persistence escalation (level 2) is the handoff: a Q2 burn or Q4 acknowledgment moves inventory out of level 1 and into K's stock, and the shadow term drops accordingly.

Amended by D7 (2026-09-03). The seam stays; what K does with it changes. `IStabilizer` gains `position()` (the signed fill q_i / cap_i), `positionCap()` and a `stale()` flag; the observer offers two aggregations, S (D4's price units) and V (cost-weighted fill, no D); the controller carries a second loop on the aggregate with its OWN gains and one integrator; and each level-1 stabilizer gets a local skew κ_i on its own band, so the distribution of inventory among the facilities has its own levers and K mid-ranges only the common mode. "K consumes positions and saturation, never filtered signals" stands. The third bullet above (no priority logic) stands too: the local skews are not priorities, they are each facility's own inventory reverting itself.

6.4.4 The monetary Diamond

The basket is already a degenerate Diamond (`BuckBasketProRata`'s `fallback` delegate calls the venue facet over shared `BuckBasketStorage`; `BASKET-REDESIGN.md` 4). The monetary Diamond is that pattern with more facets on one storage layout; facet boundaries are today's contract boundaries, so the existing forge suites carry over facet by facet:

facet	today	reports to the observer
venue	BuckBasketUniswapV3	bvib, depth D
custody / pro-rata	BuckBasketProRata (+ stress fee)	–
ops (the desk)	BuckBasketOps	its book; its bounds' saturation
undertakings + ladder	new	issued (strong side); absorbed and rho (weak side)
facility registry	new	sum of drawn credit of registered facility addresses
house	new (only if 3f wins)	OI, net position
observer	new	shadowValueInBuck() , saturation()
controller	BuckKControllerBase PID	exposes IBuckK to Buck

Buck stays the ERC-20 Diamond the master document already plans (identity, demurrage, mint/burn, ERC-20 facets); **BuckCredit** stays a sibling for the selector-collision reason the master document gives; the two Diamonds talk only through **IBuckK**, **mintFromBasket** / **burnFromBasket**, and identity. The one-way call graph survives: the controller facet reads its own storage and never calls **Buck**.

Two cautions. A single Diamond concentrates upgrade authority, so facet replacement and the ϵ / p / δ / ϕ / λ / γ parameters want time-locks – the "technical measures" discipline of D3. And the facility registry is a place an attacker could try to move K: reading drawn credit from **Buck** removes the REPORTING trust, but under D8 drawn credit itself can be fictitious until insurers are vetted, so λ_c weights only facilities backed by vetted credit.

6.4.5 What changes in wave 3

Only the seam, and cheaply. WP-3a's **shadowValueInBuck()** is the observer's output rather than a one-off view on the ops shell, with **IStabilizer** implemented first by the desk and second by the sim's agent-booked offset (a pseudo-stabilizer the **UndertakingAgent** and **FacilityAgent** write through a sim-only setter), so the WP-2/6/7 agents' books map onto facets one-to-one later. The controller stays a separate contract through wave 3 (it reads the observer); consolidation into the monetary Diamond is a post-wave-3 work package (WP-11) with the immutable-buckK decision recorded now. No wave-3 cell changes.

6.5 What this supersedes in earlier documents

- **REBALANCING-EQ.org** recommendation 2 (buy back rail margin for the growth arms): the strong-side undertaking is the supply valve the admission frontier lacked; the "top-up borrower" remains valuable but is no longer the only lever. Recommendation 4 (the oversupply flood) is now ALSO the D8-facility question (9.1). Recommendation 5 (fence-in-the-loop) narrows to the depositor stock. Roadmap item 7 ("fold the winning knobs into the BuckBasketOps/Fence revision") is realized as D3-D4-D6.
- **alberta-buck-operations.org**: "the desk may not fund from depositor TOKEN" is superseded by D2 (depositors are the wall, paid, with a fair exit); the three bounds remain and scale with the shadow deviation (9.4); **monetaryDeviationOffset** is promoted from a view to the controller's input.
- v1.0 of this document: options C and F replaced by D3 and D6; D2 (the perp) rewritten as the funding lever; the K-gradient gate cells replaced by the λ sweep.

7 The model

7.1 Symbols and conventions

symbol	meaning
bvib	basketValueInBuck: BUCK cost of one bundle (TWAP 600 s over the basket pools)
e	controller error, $1 - \text{bvib_shadow}$ (positive: BUCK dear; negative: BUCK cheap)
K	the LTV multiplier; $\text{creditLimit} = \text{activatedValue} \times K$; rails $[0, 0.95]$
τ_I	the controller's integral time (~ 90 d); the correction horizon
eps	the undertakings' bundle premium/discount; the band half-width in bundle units
δ, ϕ	the weak-side ladder: discount step per tranche; tranche size as a fraction of remaining reserve
ρ	weak-side reserve fraction remaining, R / R_0
p	the convenience factor on single-token undertakings ($\sim 2\times$ route fees)
x, y	the facility's issue (premium) and retire (discount) thresholds
D	reference depth for the shadow term (BUCK reserve of the basket pools)
I_d, I_c	net inventory of the desk and the facility: absorbed - issued (BUCK)
$\lambda_{\{d,c\}}$	shadow gains
u	facility utilization: drawn / limit over participants
R_w	weak-side reserve: NAV (TOKEN) deployable at par - eps
C_{leg}	per-leg conversion capacity: $\text{maxLegBp} \times \text{pool BUCK reserve per epoch}$
S	attack size (BUCK)
f	perp funding rate (per day), $f = g(e)$

7.2 The plant, reduced

The ops doc's `monetary_ops.py` abstraction is adequate here: one common-mode price with constant-product impact against depth and an OU reversion whose rate is what the private cast and K supply. A flow F (BUCK sold, positive) moves bvib by $\sim F/D$ at impact; the cast reverts it on its threshold-gated timescales; K moves supply through creditLimit on τ_I . Nothing below depends on the reduced form except the qualitative claims; the catalogue runs the real contracts.

7.3 The controller and the shadow term

$\text{bvib_shadow} = \text{bvib} + \lambda_d * I_d / D + \lambda_c * I_c / D$
 $e = 1 - \text{bvib_shadow}$
 $dK/dt = K_p * de/dt + K_i * e$ (conditional integration at the rails)

Signs: BUCK absorbed under the weak side (or retired by the facility) makes I positive, raises bvib_shadow, makes e more negative, and K falls harder – the contraction the absorption was betting on. BUCK issued under the strong side (or by the facility) makes I negative, lowers bvib_shadow, makes e more positive, and K rises – restoring the headroom the fast actuators spent. A Q2 burn removes the inventory from I_d ; the shadow term drops; K sees the real bvib; no double count. $\lambda \rightarrow 0$ is today's controller; λ large makes K chase every temporary operation. The mid-ranging claim: with λ at its knee, the desk's inventory returns to zero within $\sim \tau_I$ of any excursion that K can correct, and grows without bound only when the deviation is structural – which is the signal the persistence test uses to escalate to Q2/Q4.

7.4 The position loop (D7)

The two-loop form, with s in either the S or the V units of D7:

$s_S = \sum_i \lambda_i I_i / D$
 $s_V = \sum_i w_i I_i / \text{cap}_i$
 $e = 1 - \text{bvib}$
 $dK/dt = K_p de/dt + K_i e + K_q ds/dt + K_{qi} s + K_{qd} d^2s/dt^2$ (conditional integration at the rails)

Signs are D4's: absorbed inventory ($I > 0$) drives K DOWN through K_q and K_{qi} (the contraction the absorption was betting on); issued inventory drives K up. With $(K_q, K_{qi}, K_{qd}) = (K_p, K_i, K_d)$ and $s = s_S$ this is 7.3's law exactly. The plant from K to s is INTEGRATING: a change in K moves the net flow F through creditLimit on τ_I , and the active facility absorbs F while it holds price, so $ds/dt \sim F / \text{cap}$ while it is unsaturated. The position loop is therefore LEVEL

CONTROL of a surge tank fed by the demand flow and drained by K, and its tuning rules are the SIMC integrating-process rules: choose the loop's closed-loop time constant τ_s (slower than τ_I , faster than the persistence horizon), then $K_q \sim 1 / (k_s \tau_s)$ and $K_{qi} = K_q / (4 \tau_s)$, with k_s the flow per unit K per unit cap – the least-known number, to be identified in the sim (WP-16) and confirmed on the grid. The ds/dt term is inflow feedforward; it may be zero.

The mid-ranging claim restated for D7: with the position loop tuned, the aggregate position returns to within 10% of zero inside $1.5 \tau_s$ after any excursion K can correct, with at most one sign change (no limit cycle from the two integrators), and grows without bound only when the deviation is structural – the level-2 signal. The local skews (κ_i) redistribute inventory among the facilities without moving K: under a differential disturbance (one facility loaded, the level's aggregate near zero) K holds and the loaded facility's own skew unwinds it.

7.5 The undertakings

Strong side: the basket sells 1 BUCK for $(1 + \epsilon)$ bundles' worth of TOKEN (bundle path) or $(1 + p^*\epsilon)$ bundles' worth of any convertible token, converting atomically. Whenever the pools price BUCK above $(1 + p^*\epsilon)$ bundles – $b_{vib} < 1/(1 + p^*\epsilon)$ – an arbitrage exists: buy BUCK from the basket, sell it in the pools. So

$b_{vib} \geq 1 / (1 + p^*\epsilon)$ always, up to conversion capacity,

and $\geq 1/(1+\epsilon)$ with no capacity bound at all on the bundle path. Weak side: the basket buys 1 BUCK for $(1 - \epsilon)$ bundles; whenever the pools price BUCK below $(1 - p^*\epsilon)$ bundles the arbitrage runs the other way:

$b_{vib} \leq 1 / (1 - p^*\epsilon)$ while $R_w > 0$, up to conversion capacity.

The band is asymmetric in what it is backed by: issuance (unbounded) on the strong side; NAV on the weak side. The monetary book records issued and absorbed BUCK; the persistence test escalates a standing book to outright (Q2 burn of absorbed BUCK; Q4 acknowledgment of issued BUCK as permanent supply, K then carrying it) exactly as the desk does today.

The weak-side ladder. Let R_0 be the weak-side reserve in bundle units and $\rho = R / R_0$ the fraction remaining. Tranche k ($k = 1, 2, \dots$) bids for BUCK at $(1 - \epsilon - (k-1)\delta)$ bundles per BUCK and is sized at $\phi R_{\{k-1\}}$, where $R_{\{k-1\}}$ is the reserve remaining when it is struck. After n filled tranches

$R_n = R_0 (1 - \phi)^n$ (never zero at finite n) $b_n = 1 - \epsilon - n\delta$ (the next bid, bundles per BUCK) $b_{vib} \leq 1 / b_n$ (the deviation the ladder permits) $S_n = R_0 (1 - (1 - \phi)^n) / b_{avg}$ (cumulative BUCK absorbed)

so the deviation an attacker can force grows only as the LOGARITHM of the reserve fraction spent, while its average sale price worsens linearly in n . Limits: $\delta = 0$ is the flat edge (finite reserve at one price – the cliff); δ/ϕ held fixed as both $\rightarrow 0$ is the constant-product bid, whose price-versus-size curve is the basket's existing full-range position; the stableswap family interpolates. Two couplings to K follow. As tranches fill, b_{vib} rises past $1/(1 - \epsilon)$ into the sloped region – the raw error grows – AND the absorbed inventory I_d grows, raising b_{vib_shadow} ; both push K to contract, so reserve depletion is what ACCELERATES the structural correction rather than what precedes its failure. λ_d is therefore tuned with the ladder present (3c before 3d). The strong side's convenience path uses the same rule against conversion capacity: p_k rises as the epoch's C_{leg} is consumed; the bundle path is flat because it is unbounded.

The per-leg bound C_{leg} is a liveness constraint (the redemption guard must never brick), not a policy bound; it is the reason a one-day squeeze larger than the day's conversion capacity still leaves a transient trough, and the reason the directional dynamic fee (3b) and the perp's funding (3f) exist.

7.6 The attacker's cost, before and after

Today (squeeze of size S in the BUCK/USDC pool): cost = slippage against the pool's depth on the way in and out + fees, paid to the pool's LPs and the arbs; damage unbounded until the cast and K respond; at $x4$ undefended, damage lasts a year and the whale ends flat.

Under the ladder: the pool price is capped at $\text{par} + p^*\text{eps}$ by the undertaking, so beyond the band the whale buys from the basket; cost $\geq S * (p^*\text{eps entry} + p^*\text{eps exit}) + \text{fees} + f * \text{duration}$ (if the position is expressed or hedged in the perp). Damage: $|\text{bvib} - 1| \leq p^*\text{eps}$ outside the conversion-capacity window; inside it ($S > C_{\text{leg}}$ per epoch), the transient is bounded by what the cast and the directional fee price at tick speed. The harvest per attack is small and certain $\sim 2 p \text{ eps } S$ – and accrues to the single receipt class; the attacker's expected value is negative at every size, so the rational attacker does not attack. This is the target-zone result: the band converts the harvest into deterrence.

7.7 Reserve adequacy on the weak side

The weak side is the reserve-bounded flank. With a flat edge its credibility condition would be

$R_w \geq \max \text{ over plausible scenarios of integral over } \tau_I \text{ of (net BUCK sold to the basket at } \text{par} - \text{eps})$

– the reserve must outlast K 's correction of the deviation producing the outflow, and the attack is timed to the day it does not. With the ladder the condition changes character: the reserve is never exhausted at a finite price, so adequacy becomes a statement about the DEVIATION the system tolerates per fraction of reserve spent – the choice of (δ, ϕ) – and about whether K 's correction arrives before the deviation the ladder permits at the plausible cumulative outflow exceeds the tail gate. Formally: for the worst plausible outflow S over τ_I , the ladder's $n(S)$ satisfies $1 / (1 - \text{eps} - n(S) \delta)$ within the deviation the gates allow, with the shadow-coupled K restoring ρ toward 1 inside τ_I . The catalogue's $x4$ grind-down with the full cast (parity held, K 0.936, 17% railed) is the empirical frontier at today's NAV with an implicit CPMM ladder; wave 3c re-measures it across (δ, ϕ) and wave 3h with the seeder. Sources of R_w beyond NAV: the desk's capitalization grant and treasury accrual (existing), and the cat-bond vault (D6, optional). Nothing else.

7.8 The facility

Participant i has limit L_i , drawn d_i , thresholds (x_i, y_i) , and a deployment cap m_i . Hazard of issuing rises with the premium beyond x_i and with headroom $(L_i - d_i)$; hazard of retiring rises with the discount beyond y_i and with d_i . Aggregate utilization $u = \sum d_i / \sum L_i$ is a fast actuator's position and enters I_c . Pro-cyclicality (4.6) is bounded by K : a premium regime raises K and headroom, so the facility can issue more into the premium – which is the correct direction – and a discount regime lowers K , which forces retirement on holders near their limit, again the correct direction. The structural risk is D8: a participant whose "insured asset" is fictitious issues free BUCK into every premium with no retirement obligation that bites (9.1).

7.9 The perp funding lever

A perpetual settled on a medianized bvib TWAP, with funding $f = g(e)$ paid continuously by the side positioned AGAINST par (longs- bvib pay when $\text{bvib} > 1$; shorts- bvib pay when $\text{bvib} < 1$) to the house, which is the monetary book. Because the band bounds bvib , the perp's payoff is bounded, and its equilibrium funding is the market's price for holding a band-break bet – the currency board's automatic interest-rate mechanism applied to speculators only. Solvency condition for listing: for

the defended excursion envelope measured in wave 2 (peak $\times$ duration at $x1..x4$), the funding + eps + fee surge an attacker pays to express the raid through the perp must exceed the payout the bounded excursion can return; OI is capped against the defense budget. If it does not price safely, it is not listed.

7.10 Predictions the catalogue can falsify

1. **Band integrity.** With the strong-side undertaking at $\text{eps} = 3\%$, $p = 2$, the $x4$ squeeze (\$40M in a day) never takes bvib below $1/(1 + p \cdot \text{eps}) \sim 0.943$ beyond the conversion-capacity window, and K ends interior (< 0.90). Today: 0.69 and 0.95 railed.
2. **Depositor fairness.** With the stress fee, depositor realized P&L in every $x4$ cell is $\geq$ its value in the $x1$ cell of the same arm. Today: $-\$1.99\text{M} / -\2.72M .
3. **Attacker cost.** Whale P&L $\leq -2 p \text{ eps } S$ on every undertaking-bounded squeeze; the whale never ends flat on a dump (today: $\$0.00\text{M}$ at $x4$).
4. **Mid-ranging.** With λ_d at its knee, the desk's inventory returns to within 10% of zero inside $1.5 \tau_I$ after every $x1/x2$ injection, and K 's rail margin at $x4$ is at least that of the "all" mix today ($K \text{ end} \leq 0.81$). With $\lambda_d = 0$ the desk's inventory grows monotonically on the drift arm (runaway one).
5. **Benign-flow health.** Quiet-regime tracking error of every basket pool against the truth-implied price is unchanged within $x1.25$ under the undertakings, the seeder, and the directional fee.
6. **The middle of the ladder.** A facility population scaled to the "plentiful un-leveraged asset base" cuts the $x1$ squeeze peak and AUC below the "credit" mix's and REALIZES (not merely marks) its harvest on the retire leg.
7. **Duration pricing.** With the perp, the whale's accumulation-phase cost rises with the days it takes, and the detector fires at least 20 days before the dump with a false-positive rate on the control below one event per two years.
8. **The ladder has no cliff.** Across the (δ, ϕ) grid on the $x4$ dump and grind-down, no cell shows the weak-side reserve fraction ρ below 0.1 while bvib stays inside the first edge; the whale's average sale price worsens monotonically with cumulative size in every cell; and the ladder cells reach a LOWER peak deviation per dollar of reserve spent than the flat-edge ($\delta = 0$) cell at the same eps .
9. **Habituation (D7).** With the position loop tuned (WP-16), the level's aggregate position $|s|$ returns to within 10% of zero inside $1.5 \tau_s$ after every disruption of classes 1, 2 and 5 that K can correct, with at most one sign change (no limit cycle between the two integrators), under both S and V ; and under V the result is unchanged by the class-10 liquidity withdrawal, where S 's D-scaled response shifts.
10. **Book-loading cost (D7).** An attacker who loads a facility's inventory to move K and unwinds after the move (class 9) ends with P&L ≤ 0 under both S and V , and K 's excursion per unit of loaded fraction under V is no larger than under S .

11. **One integrator (D9).** Under S0k the two integrators' drift (the total variation of `pid_i - pid_qi`, decision 31) is nil by construction and K shows at most one sign change per disruption of classes 1, 2 and 5; under S and V the drift is non-zero on every arm, grows with the number of absorb-unwind cycles, and some S or V cell holds an integrator at its rail while K is interior. If S0k is not also within the gates' noise of S and V on G1-G8, the position P term matters and stage 1's second outcome applies.
12. **The feedforward's worth (D9).** On the dump x4, grind and cohort-step arms the ordering `A <= S0k <= estimator <= ceiling` holds on peak, AUC and the book's BUCK-days; the ceiling beats S0k by more than the gates' noise on BUCK-days and habituation time and by little on peak, because the feedforward sits at the rail during the disruption and earns its keep on the release. A ceiling within the noise of A falsifies K as the lever for these disturbances (stage 2's first outcome).

Any of these failing changes a decision; 9.6 says which.

8 The wave-3 program

The catalogue's rule applies throughout: no new machinery before the falsifiable instrument, and no contract work before a catalogue win. Cells are `catalogue.py` grid entries (arm x mix x scale, plus `--set` overrides); new physics arrives as agents first and contract features second, because the sim runs the real contracts and an agent with the right authority (SimLP's zero-premium credit and V3 callback proxy) can stand in for a contract feature at the economics level before the Solidity exists.

8.1 Instrument: the markout / LVR ledger, with the basket-side columns (3a)

Definitions. For each swap `s` in constituent pool `i` at time `t`, with the basket-LP side's signed base-token flow `q_lp` and execution price `p_exec`:

$$\text{markout_D}(s) = q_lp \times (p_ref(t + D) - p_exec(t))$$

Reference for the COMMON-MODE ledger: PAR (reversion to 1 at horizon D). For the DIFFERENTIAL ledger: the truth-implied TOKEN/BUCK, and (sim-only) the exogenous reference series. Horizons `D` in {next tick, 1 d, 5 d}. Per pool per window: `fees_i`, `adverse_i` (sum of markouts), `netlp_i` = `fees_i` + `adverse_i`, `lvr_bench_i` (differential only), `carry_ratio_i` = `fees_i` / `max(lvr_bench_i, -adverse_i)`.

Attribution. Every sim swap has a known sender class: whale (accumulate / dump / re-buy separately), BUCK-unaware arbs, director keeper, excursion agents by base, DM depositors, debtors, and – new – the undertakings, the facility, the seeder, the house. Toxicity of a class = mean markout per dollar of its flow. Expected: whale dump-day flow deeply negative for the basket; benign arbs mildly negative gross, ~zero net of fee; the keeper netting to ~fees; excursion agents negative for the pool ledger by design (the defense partly funded by the basket's own book – confirmed in outline by the x1 dump: defenders +\$0.26M, basket NAV -\$0.75M).

Where it lives. `snapshot.py`: swap-event capture with class tags and deferred markout at `t + D`; `basketNav` / `treasuryBuck` / `dmProfitUsd` promoted into the catalogue summary. `eqmetrics.py`: a markout panel beside `excursion_windows` / `excursion_response`. `catalogue.py`: grid columns – basket fees, adverse, netlp, carry_ratio (worst pool), NAV USD and NAV-in-baskets delta vs undefended, treasury delta, depositor realized delta, whale toxicity.

Cells. The existing 86 cells, re-summarized from their vectors (`--report-only` after the metric lands); no new physics.

Gate. None new for 3a; it is the instrument. Deliverable: the table in 3.3 produced by `catalogue.py` rather than by hand.

8.2 Gates for every later wave

1. The 2-year tail gate (`eqmetrics.accept`): tail bvib within 1.00 ± 0.02 , K never railed in the tail, channels alive.
2. **Carry coverage:** tail `carry_ratio_i` ≥ 1 for every constituent pool.
3. **Benign-flow health:** quiet-regime tracking error of each pool vs the truth-implied price within baseline $\times 1.25$.
4. **Duress-exit fairness:** depositor realized P&L in a defended or undertaking cell $\geq$ the undefended cell of the same arm; the transfer between exiting and remaining depositors has the remaining side positive.
5. **Band integrity** (from 3c on): bvib within $[1/(1+p*\text{eps}), 1/(1-p*\text{eps})]$ at every frame outside the conversion-capacity window of an injection; the undertakings' books return to zero within $1.5 \tau_I$.
6. **Attacker cost:** whale P&L $\leq -2 p \text{ eps } S$ on bounded squeezes; never flat on a dump.

8.3 The waves

wave	axis	cells
3a	instrument	existing 86 cells, re-summarized
3c	the undertakings	$\text{eps} \{2,3,5\} \times p \{1.5,2,3\} \times \{\text{dump, squeeze, grind-down, grind-up}\} \times \{x1, x4\}$; undefended AND full
3d	shadow-bvib controller	$\text{lambda_d} \{0, 0.25, 0.5, 1, 2\}$ with <code>--basket ops</code> ENABLED in the loop (first time), on dump/squeeze
3b	directional dynamic fee	$\{\text{static-retune, directional-RV, directional-size}\} \times \{\text{dump, squeeze}\} \times \{x1, x4\}$
3g	the facility	$\text{population} \{8, 32, 128\} \times (x, y) \{(2\%,2\%), (3\%,3\%), (5\%,2\%)\} \times \{\text{deposit-to-basket on/off, on squeeze}\}$
3f	perp funding lever	$g() \text{ slope} \{0, 1, 3, 10\} \times \{\text{whale-via-perp vs whale-via-spot, on squeeze } x1/x4 \text{ and the accumulation ph}\}$
3e	derivatives on satellites	D1 hedge on/off $\times \{\text{spike, step}\} \times \{x1.3, x1.6\}$
3h	seeder, seeds, floods	seeder on/off at launch; the campaign's 5 seeds on the key 3c/3d cells; the oversupply flood (rec. 4) w
3i	the controller alternatives (D7)	the WAVE3.org test plan: 10 disruption classes $\times$ mixes $\times$ capacity $\times \{0.25 \dots 16\} \times$ designs $\{A, S0, S,$

Order: 3a -> 3c -> 3d -> 3b -> 3g -> 3f -> 3e/3h. 3c precedes 3d because the undertakings change the desk's book size, which is what the lambda sweep must be measured against; 3g precedes 3f because the middle of the ladder decides how much duration pricing is left to do. Amended 2026-09-03 (D7): 3d's lambda sweep is absorbed into 3i, which runs after decision 8 and BEFORE 3c's verdicts are read, because every "K end" number depends on which K is in the loop.

8.4 Implementation notes for the sim

- **UndertakingAgent** is the fastest honest stand-in for D3. SimLP can mint against a zero-premium credit with an arbitrarily large face and hold both sides of a V3 position; the basket-rotation execution path already trades `TOKEN <-> BUCK` in every constituent pool in `basketAmount` proportions. Standing orders at par $\pm \text{eps}$ in bundle units are then eight range orders (one per pool) re-struck daily, or, more simply, a daily check that fires when bvib crosses the edge and trades up to `C_leg` per pool. The agent must book its issued and absorbed BUCK so 3d can read `I_d`.

- **The shadow controller** needs Solidity: `BuckKControllerDirect` reads `basket.basketValueInBuck()`; a `Shadow` variant reading a new `shadowValueInBuck()` on `BuckBasketOps` (`bvib + lambda x monetaryDeviationOffset() / depth`) is small, and the sim deploys whatever the repo compiles. Until the undertakings are contract-level, the agent's booked inventory must be pushed into the offset (a governance setter on the ops shell for the sim, removed later).
- **The facility** is the credit base with two changes: the issue leg deposits into the basket instead of selling into BUCK/USDC, and the retire leg redeems the receipt and repays. Utilization telemetry is a counter in `snapshot.py` (add it in the same commit – the ops doc's "the agent is invisible" lesson).
- **The perp** is a Python-side ledger (positions, mark = medianized bvib, funding accrual) whose funding settles on-chain as BUCK transfers between the whale's proxy and the house's proxy each tick; the whale gains a `via = "perp"` knob. No contract until 3f wins.
- **The treasury seeder** is `BuckPoolInvestorAgent` with the range struck from the current BUCK/USDC tick toward the truth-implied ideal, a withdraw rule keyed to SimLP-independent depth, and the ops doc's orientation checklist applied.
- **The directional fee** is a fee schedule in the pool adapter used by the router quoter and the sim's swap path; the v4 hook comes later behind the venue seam of `BASKET-REDESIGN.md` 4.
- **Telemetry discipline** (ops doc): counters into `snapshot.py` in the same commit as the agent; per-agent inventory through `arb_state`; several seeds before any peak-damping number is quoted.

8.5 Compute

At ~1 core-hour per 2-year pyrevm cell (the catalogue's measured rate), 3a is free, 3c is ~72 cells, 3d ~40, 3b ~24, 3g ~72, 3f ~24, 3e ~8, 3h ~40 – ~280 cells, a few days of a MATRIX_JOBS-way machine, in the order above so each wave can stop the next.

9 Risks, open questions, and what would change the decisions

9.1 D8 and the facility

The facility lets latent credit issue into premiums. A participant whose insured asset is fictitious (D8: `createCredit` is permissionless and zero-premium credit is exempt from the funding gate) issues free BUCK into every premium with a retirement obligation that never bites (no real asset, no reason to close). This is the oversupply flood of `REBALANCING-EQ.org` recommendation 4, automated. The facility must be restricted to vetted insurers' credit (the on-chain solvency-plus-rating scheme the master doc names as the mainnet blocker) or, until then, to a whitelisted insurer set in sim and pilot. 3h measures the flood with and without the facility to size the exposure.

9.2 Band-break credibility on the weak side

The weak side is NAV-bounded. Bertola-Caballero says an imperfectly credible edge destabilizes near it, and Krugman-Flood-Garber say the attack is timed to the day a flat edge stops being honored. The ladder (D3 refinement, 7.4) is the structural mitigation: there is no day on which the weak side stops being honored, only successively poorer prices, so there is no discontinuity to

time and no size to probe – the RULE (the tranche schedule against the on-chain reserve) is what is announced. The residual risk moves to the ladder’s parameters: a delta too small reproduces the cliff, a delta too large lets a modest outflow force a deep deviation. Wave 3c’s (delta, phi) grid, the perp pricing band-break bets, and a reserve visibly large relative to tau_I outflow (7.5) are the answers.

9.3 Sensor damping by the invisible cast

Private defenders damp bvib and K cannot see their books. Their capital is finite and their positions temporary by design, so the lag is bounded by their size; but a very large private cast makes K late. The shadow term could include an estimate of private inventory from flow attribution; it is deliberately left out until 3d shows the desk-only term is insufficient.

9.4 Conversion capacity and the one-day trough

The bundle path is unlimited but the convenience path and every constituent-pool leg are bounded by C_leg for liveness. A one-day squeeze larger than a day’s capacity still troughs. The directional fee is the tick-speed answer; if 3b shows it insufficient, the remaining lever is a larger per-leg bound with a redemption guard that tolerates it (a TWAP-window-aware guard rather than a flat maxLegBp).

9.5 Governance abandonment

The SNB failure mode is available to any governance that can move par. Par is definitional here and the undertakings are rules, not operations; the residual is a governance that changes eps, p, lambda or the basket under pressure. Time-locks on those parameters, and the announcement discipline of Hong Kong’s technical measures, are the answer; they are policy, not code.

9.6 What would change which decision

observation in wave 3	changes	to
band integrity fails at x4 even with unlimited strong side	D3 (eps, p)	larger eps, larger C_leg with a tolerant guard
the ladder’s peak deviation per reserve dollar is no better than the flat edge	D3 (delta, phi)	steeper or shallower schedule, or the CPMM li
depositor realized P&L still negative with the stress fee	D2	a larger fee schedule, or a redemption-reserve
desk inventory never returns to zero at any lambda	D4	K’s tau_I, or an explicit Q2 rule keyed to inve
the facility issues into premiums but never retires	D6	mandatory retire thresholds, or facility restrict
the perp raises attacker cost less than its liquidation risk costs the house	D5	do not build the perp; rely on the band and th
benign tracking error rises under the seeder or the fee	D5, 3b	narrower seeder range; lower base fee
the flood equilibrates below theta x apr with the facility on	D6 + D8	the facility waits for insurer vetting
S0k matches S / V on the gates with less drift and fewer sign changes	D7 -> D9	one PI on the shadow error; the position loop
the perfect-measurement K is within the gates’ noise of A	D9, dec. 28	K keeps the structural residual; spend on facil
the estimator’s persistence call fires on organic noise	D9 stage 3	per-channel flow estimation, labelled by source
g asymmetric beyond 2x, or the supply ramp compounds without saturating	D9 stage 3	a piecewise or population-dynamics law; never

9.7 The JS port

alberta-buck-platform.org ports only agents whose excursion control the catalogue demonstrated, in their simplest embodiments. The undertakings, the shadow controller and the stress fee are CONTRACT features and port with the contracts; the facility, the seeder and the house are agents and port only after their cells win. The perp ledger is Python-side in 3f and is not a port candidate until it is a contract.

10 Recommendations, ranked by expected information per unit of compute

1. **Build 3a**, including the basket-side columns – the table in 3.3 belongs in the summary, not in a scratchpad script.
2. **3c in the sim** – the undertakings are the largest single change to the physics and the x4 squeeze is the cell that decides them.
3. **3d** – put the desk in the loop for the first time, with the shadow term; find lambda's knee the way the 20-day knee was found.

3a. **3i (D7) before 3c's verdicts are read** – the controller under judgment changes every "K end" number; run the controller- alternatives grid (WAVE3.org) on the equilibrium backdrop first.

3b. **D9's stage 2, the perfect-measurement K, before cycle 2** – one Python cell per arm bounds the value of every controller package after it, and costs less than any of them.

1. **3b** – the directional fee, for the one-day trough.
2. **3g then 3f** – the middle of the ladder, then the duration lever.
3. **3e, 3h** – satellites, seeds, the seeder, the flood.
4. **Contract work LAST**, in this order: the undertakings and monetary-book accounting on the ops shell; `shadowValueInBuck` and the `Shadow` controller; the stress fee in `_redeem`; the facility; a v4 hook for the winning fee policy behind the venue seam; the perp only if 3f closes. Nothing ships on-chain that did not first win its catalogue cell.

11 Glossary

- **bvib**: `bucketValueInBuck`, the BUCK cost of one basket bundle; the controller's process variable; > 1 = BUCK cheap (discount), < 1 = BUCK dear (premium).
- **common mode / differential mode**: BUCK against the bundle (K's job) vs constituent against constituent (the director's job); the mean and the differences of the same tick-EMA ladders.
- **the desk**: `BuckBasketOps`; Q1 absorb, Q2 retire, Q3 supply, Q4 issue.
- **the undertakings**: the strong-side (sell BUCK, unlimited) and weak-side (buy BUCK, NAV-bounded) standing offers at par +/- eps in bundle units; the currency board.
- **the weak-side ladder**: the weak side struck as tranches at successively deeper discounts (eps + (k-1) delta), each sized at a fraction phi of the reserve remaining; no cliff, superlinear attacker cost, and a deviation that gauges reserve stress.
- **the facility**: the credit auto-pay-down facility; latent credit issuing into premiums and retiring into discounts on user-set thresholds.
- **the shadow bvib**: bvib plus the inventory the visible fast actuators have absorbed minus issued, scaled; what K sees.

- **the seeder:** the treasury’s temporary BUCK/USDC position from the current price toward the ideal.
- **the house:** the monetary book as counterparty and funding recipient of the bvib perp.
- **the cast:** the private excursion agents of the catalogue (usdc, buck, credit, basket, crb).
- **manufactured reversion:** the property that par is definitional and K plus the cast force bvib back to it – what makes selling common-mode convexity sound.
- **the position loop:** K’s second loop, on the level’s aggregate position s (in S or V units), with its own gains (Kq , Kqi , Kqd) and one integrator (D7).
- **S / V:** the two aggregations of the stabilizers’ books under judgment – the shadow bvib in price units ($D4; \sum \lambda_i q_i / D$) and the cost-weighted fill in $[-1, 1]$ ($\sum w_i q_i / \text{cap}_i$).
- **the local skew:** $\kappa_i q_i / \text{cap}_i$ added to a facility’s own band, so its own flow reverts its own book; the differential-mode lever (D7).
- **habituation:** the return of the fast actuators’ inventory to zero as the slow lever takes over the steady-state load (Henson et al. 1995); measured as the time for $|s|$ to fall within 10% of zero.
- **the shadow error:** $e_s = e + \lambda s$, the one signal K integrates under D9 – the price term live while no facility is engaged, the position term live while one is; S0k’s input as built.
- **the flow estimator:** the alpha-beta (steady-state Kalman) filter on the fused net BUCK flow through every peg-supporting channel, split into a persistent and a transient part; the sensor of D9’s feedforward; level 2 of 6.4 made explicit.
- **the perfect-measurement K:** D9’s ceiling – the feedforward law run with the sim’s true flows and a known g ; the bound no estimator can beat, and the measure of whether K is the lever at all.
- **the uptake lag (τ_K):** the time the supply takes to answer a K move through the borrowers’ hazard; with g it sets the release timing (the supply still in the pipe).
- **the drift:** the total variation of $\text{pid}_i - \text{pid}_{qi}$, the two integrators’ contents shifting between them without moving K (WP-16 finding 4; WAVE3.org decision 31).

12 References

12.1 Repository documents and code

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- alberta-buck-rebalance.org; BASKET-REBALANCE.md; src/basket/PairsRebalanceDirector.sol, BasketRebalanceDirector.sol, RebalanceDirectorBase.sol.

- `BASKET-REDESIGN.md`; `src/basket/BuckBasketProRata.sol`, `BuckBasketUniswapV3.sol`, `BuckBasketStorage.sol`, `BuckBasketFence.sol`.
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12.2 The critique

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